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QuantumNeuroXAI: A Quantum-Inspired Deep Learning Framework with Explainability for Brain Signal Analysis and Neurological Disorder Detection

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Abstract

Electroencephalography (EEG) is a non-invasive, high-temporal-resolution method for diagnosing and monitoring neurological disorders. Deep learning has recently substantially enhanced the state of the art for automated EEG analysis. However, many of the currently applied paradigms are still challenged by limited generalisation across datasets, vulnerability to noise or preprocessing changes, and the absence of interpretable decision rules. Additionally, many deep learning models operate like black boxes, which limits their use in clinical settings where interpretability and trust are key. Although the potential of quantum-inspired learning has recently been demonstrated through improved feature separability in high-dimensional signal spaces, its scope of applicability does not yet extend to deep temporal modelling and explainable artificial intelligence applications. We address these limitations by introducing QuantumNeuroXAI, a quantum-inspired deep learning framework implemented on

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classical hardware that leverages structured feature encoding inspired by quantum neural networks to provide inherent explainability for EEG-based diagnosis of neurological disorders. This framework hybridises quantum-inspired feature encoding with a deep-learning architecture that blends temporal convolutional and attention-based recurrent modelling to capture local and long-range patterns of dependencies in EEG signals. The framework incorporates a multi-level explainability module relevant at the signal, model, and quantum-representation levels, allowing predictions to be interpreted in a clinically meaningful and transparent fashion. We conduct extensive experiments on three publicly available EEG datasets (TUH EEG, CHB-MIT, and BCI Competition IV-2a) to evaluate the proposed framework. These quantitative results show that QuantumNeuroXAI achieves statistically significant and large effect sizes, with macro-F1 improvements of up to 5.2% over classical machine learning, deep learning, and hybrid baseline models. Additional robustness and scalability analyses further validate stable performance against dataset shift and across various preprocessing configurations. In summary, QuantumNeuroXAI is an interpretable and reproducible solution for EEG-based neurological analysis, demonstrating promise for clinical decision support and future scalability to multimodal brain signal applications. It is important to note that the proposed framework does not rely on quantum hardware and is fully implemented using classical computational resources. The implementation of the proposed framework is publicly available at: https://github.com/venkateshwarlu-bondu/QuantumNeuroXAI

Keywords - Electroencephalography, Quantum-Inspired Learning, Deep Neural Networks, Explainable Artificial Intelligence, Neurological Disorder Detection

1. INTRODUCTION

Due to its high temporal resolution, portability, and cost-efficiency, electroencephalography (EEG) is still one of the most common non-invasive techniques for monitoring brain activity and aiding the diagnosis of neurological disorders. Despite the increasing prevalence of conditions, such as epilepsy, motor impairments, and cognitive dysfunctions, and the demand for accurate and timely decision-making support, automated EEG analysis systems remain underutilized. Traditional EEG analysis pipelines were based on handcrafted features in the time, frequency, and time–frequency domains, followed by classical machine learning classifiers. Although such methods have shown satisfactory performance, they tend to struggle to learn the complex, nonlinear, and highly time-variant spatial-temporal patterns underlying features, as there is significant inter-subject and cross-dataset variability in multi-channel EEG signals [1], [2].

Since deep learning has significantly simplified EEG-based classification and prediction through end-to-end representation learning using convolutional, recurrent, and attention-based architectures [3], [4], EEGLP is mainly focused on extracting features from recorded EEG data. These models have outperformed traditional methods and are capable of automatically learning in-depth discriminative features from raw or minimally preprocessed EEG data. That being said, the state of the art is not free from several limitations. However, several deep learning-based methods are not robust to dataset shift, sensitive to noise and various preprocessing choices, and mostly black-box models with limited interpretability [5], [6]. Such opacity is especially problematic in clinical contexts where explainability is critical for trust, validation, and regulatory acceptance of model predictions [6]. In addition, recent quantum-inspired learning methods have introduced potential enhancements to feature separability in high-dimensional signal spaces [7]. However, in existing works, these ideas are commonly used in isolation and not integrated into deep temporal architectures, nor systematically analysed for explainability [8].

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This set of challenges emphasises the need for a holistic framework that can simultaneously tackle expressive representation learning, robustness, and the interpretability of EEG-based neurological analysis. This gap motivates the current research, which seeks to build an integrated approach that combines the strengths of quantum-inspired learning and deep neural networks (DNNs) whilst ensuring explainability as a design principle rather than an afterthought. The main aim of this work is to develop and validate a quantum–deep learning framework for classification of EEG signals into normal and abnormal, that holds the capability of (1) accurately detecting neurological disorders, (2) generalizability across heterogeneous datasets, (3) providing multi-level explanations that are consistent with neurophysiological understanding.

While the proposed QuantumNeuroXAI framework integrates existing technologies by combining multi-modal data integration with cross-modal knowledge transfer, it still offers several novel contributions. It introduces a quantum-inspired feature-encoding strategy that projects high-dimensional EEG representations into a lower-dimensional space, improving class separability and facilitating more efficient, robust learning. Then it combines this encoding with a hybrid deep learning architecture that employs temporal convolution and attention-based recurrent modelling, enabling both local and long-range temporal dependence modeling in EEG signals. Third, the framework proposes a hierarchical explainability design at the signal level (channels, time, and frequency), the model level (attention and feature attribution), and the quantum-representation level (feature sensitivity) to achieve transparent and clinically interpretable outcomes. Furthermore, we have performed a systematic evaluation of the framework using only locked public EEG datasets, enabling reproducible assessments, fair comparisons, and the widespread validation of the proposed framework.

It is important to clarify that the proposed framework is based on *quantum-inspired* learning principles rather than actual quantum computing hardware. All quantum-inspired transformations are implemented using classical simulations, ensuring scalability, accessibility, and practical deployment without requiring specialized quantum devices.

This paper makes the following contributions to the research literature: a quantum deep learning architecture for EEG-based detection of neurological disorders. Across five public datasets, it shows statistically significant and practical improvements over classical, deep, and hybrid baselines. It presents a holistic explainability evaluation strategy that assesses not only visualisation but also consistency and stability. Third, there is an extensive experimental evaluation of robustness, scalability, and statistical significance, which reinforces the reliability of the proposed method for practical use.

The rest of this paper is structured as follows: Section 2 includes related work on EEG-based neurological analysis, deep learning models, quantum-inspired methods, and explainable AI. In Section 3, we present the preliminaries, including EEG signal modelling, problem formulation, andbasic concepts of quantum-inspired learning and explainability. Section 4 explains the proposed methodological approach (QuantumNeuroXAI), dataset description, copying, model architecture, explainability design, and training protocol for the experiment. Section 5 provides results of the experiment, a comparative evaluation, an ablation study, and statistical analysis. The results are presented in Section 6, which also describes the study's limitations. Lastly, Section 7 summarises the paper and discusses future research.

2. RELATED WORK

The current section summarises previous work on machine learning, deep learning, quantum-inspired, and explainable AI methods for brain signal analysis, highlights

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methodological trends, limitations, and open research gaps in these methods, which motivate the design of the proposed QuantumNeuroXAI framework.

2.1 Broad AI Adoption for Neurological Disorder Detection from Brain Signals

Surveys have repeatedly cited the empirical craze for EEG tracking as the most scalable solution for low-cost, portable neurological screening; however, the associated practicalities (e.g., noise, non-stationarity, and cross-subject variability) can create non-trivial obstacles to implementing reliable models. Khalfallah et al. [1] et Akter Lima et al. Narayan et al. [3] summarise a survey of these classes of constraints and their influence on end-to-end pipelines. According to [3], however, the majority of ML deployments still fail to demonstrate clinical-grade decision-support correctness from lab-grade performance. Similarly, neuro-AI review articles claim the field is advancing towards hybrid schemes that combine engineered priors (which tend to be more interpretable) with representation learning (which often excels at stability) to engineer solutions capable of operating in real-world environments (Raghavendra et al., 2023). [6]; Surianarayanan et al. [9]; Bhatele and Bhadauria [8]).

2.2 Classical Feature Engineering and Traditional ML: Still Strong Baselines

The recent literature on EEG diagnosis still essentially treats compact ML (signal processing + compact ML) as a competitive baseline, particularly when datasets are small or computational limits are significant. Khosla et al. Alturki et al. [12] rather summarize comparative evidence across time–frequency representations and classifiers. For recognition of neurological disorders, DWT-bandpower features and SVM-style decision boundaries are shown in [10]. Likewise, Shoieb and Youssef [11] mention that computer-aided detection approaches remain appealing because they can be easily interpreted and deployed. Frequency-band decile strategies, presented in [7] by Fernandez and López, illustrate additional representational choices that may help realise strong separability without intense networks. Such lines of work spur modern hybrids: irrespective of how successful deep learning has become, well-engineered baselines from scratch are crucial for any comparative claim, as well as for claims of novelty driven by ablation studies.

2.3 Deep Learning in EEG-Va Based Neurological and Psychiatric Disorders

Recent advances in deep learning have expedited EEG-based screening across epilepsy, Alzheimer's disease (AD), schizophrenia, depression, attention deficits (ADHD), and more general psychiatric conditions. Jo et al. Deep learning–based diagnosis of neurological disorders has been expanded from EEG [13] to magnetoencephalography (MEG) signals and to a deep neural network architecture [14]. The MEG: deep learning study showed that MEG, combined with end-to-end deep representation learning, enables the identification of subtle spatiotemporal neural patterns related to neurological diseases. The findings underscore the potential of deep learning to improve the analysis of non-EEG brain signal modalities while supporting the generalizability of neural network–based approaches across heterogeneous brain signal sources. The expansion of the field is evident in systematic reviews but also highlights the fragmentation of datasets and non-standardised validation processes. An approach-oriented, routine style characterises method sections of COVID-19 prediction research: a systematic review (Uyanik et al.), de Bardeci et al. [14]; Kensari et al. [15]; Nysa et al. [16]; Jain et al. [17]; Gautam and Sharma [18]; Gautam, R et al. [19]; Bardeci et al. [23]; Rivera et al. [24]; Ahmed et al. [22]). Model-wise, Ruffini et al. While Oh et al.[25]Bring CNN/RNN learning to EEG spectrograms for prodromal neurodegeneration scenarios. Case (26) is an example of a CNN for schizophrenia diagnosis with EEG. Alves et al. A later paper [21] goes beyond raw time series to learn from functional connectivity matrices with deep networks, further supporting the idea that brain organisation may be more effectively represented using a graph-like encoding. Sanchis et al. Classifying ADHD by EEG + deep learning around brain-region relevance [28]. Lastly, Altıntop [5]

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highlights the need for x-disease vs. y-disease discrimination (not only healthy-vs.-disease), a more difficult but clinically relevant scenario that frequently reveals the brittleness of a model.

2.4 Hybrid / Neuro-Fuzzy and Multimodal Brain-Signal Modelling

Outside the CNN/RNN families, paradigmatic hybrids begin leveraging EEG non-stationarity and providing informativity to improve decision transparency. One direction of representation is neuro-fuzzy learning: Dhanalakshmi Gopal et al. Adaptive neuro-fuzzy inference applied to multimodal brain-signal analysis for cognitive pattern recognition [20]. Related EEG processing evidence a long-lasting need for principled feature reduction and robust temporal modelling (Khan et al. [27]). These results relate to a broader movement towards taking a "model-informed" approach to learning, where neurophysiological plausibility, feature constraints, and modularised, structured inference are incorporated rather than used as post-hoc options. Positioning QuantumNeuroXAI as an end-to-end design that can integrate powerful representation learning whilst having explicit hooks for explainability vs treating interpretability as an afterthought is subtle, right?

2.5 Quantum Learning and Quantum-Inspired Learning for EEG Classification

There is growing interest in leveraging quantum-inspired ML to classify EEGs directly in high-dimensional space. A typical approach can, under local data scarcity, target boosting decision boundaries or feature expressiveness. Conceptual quantum-based autism detection using EEG spatial patterns has been explicitly framed by Saranya and Menaka [4] within a growing perspective in which quantum kernels or quantum neural models are believed to supplement classical feature learning when separability is subtle. Simultaneously, broader neurotechnology assessments contend that AI-aided customisation and adaptive decision-making will increasingly become cornerstones of BCI systems and digital prescriptions (Chai et al.). Because they need to be deployable safely, closed-loop brain stimulation systems require not only reliable sensing and modelling (Belkacem et al. [16]), but also robust control. Seen through this prism, QuantumNeuroXAI is in step with an actionable timeline: quantum-inspired learning is not just academic, newsworthy, but a real-world candidate block in a looped neuro-AI stack–– if explainability and validation are treated productively.

Hashem et al. An integrated machine learning based BCI framework for classification of various limb motor tasks from EEG signals [50]. One of the main contributions of this work is its explainable modelling strategy, which enables clinicians to gain insight into which EEG features contributed most to motor task predictions. The third study explores BCI interpretability, which is a critical requirement for high-impact rehabilitation and assistive neurotechnology applications where trust and transparency are crucial; highlight link to comment.

2.6 Explainable AI for EEG: From Post-Hoc Maps to Clinically Actionable Insights

The “black-box” nature of many EEG deep models is indeed one of their main drawbacks, as it lowers the trust of clinicians and makes error analysis more difficult. Akbar et al. NeuroFusionNet [30] is a hybrid EEG feature fusion framework for accurate and explainable detection of Alzheimer’s disease. They combine multi-domain EEG features with deep learning and utilise SHAP-based explainability to determine clinically significant spectral and spatial biomarkers. These results reveal the potential of feature-level fusion with interpretable machine learning to improve the robustness of diagnostic predictions in a cognitively believable manner, precisely matching the principles of transparent decision-support systems for neurologists. Shahtalebi et al. Earlier works, such as the deep, explainable framework for discrimination of neurological disorders by [29] and the work by Woodbright et al. Feature relevance analysis and transparent

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design choices are necessary building blocks for trustworthy neurological AI [36]. Presacan et al. [32] and Bibi et al. Synthesising recent XAI advances in EEG and brain-disease pipelines, [31] point out that explanation approaches need to have neurophysiological grounding and be cross-validated against stable explanations, rather than pleasing patchworks of false-colour heatmaps. Task-specific explainable EEGs are also growing: Li et al. It is possible to identify task-based EEGs as a biomarker for preclinical AD by applying explainable deep learning [33] and the Khaleghi et al. deep learning study [52] for detecting depression from EEG. Islam et al. found that in domain-specific settings. We propose an explainable EEG framework for stroke prediction [44] and Hussain et al. LIME-style explanations are adequate for EEG-based activity recognition [48], and the increased portability of local explanation tools suggests they could also be viable for other applications. On the methodological level, newer pipelines are being used that embed SHAP/LIME into the evaluation itself [56] for epilepsy detection, Mouazen et al. SHAP-based autoencoder + Bi-LSTM for early detection of neurodegenerative disease. [39]

2.7 Explainable Deep Learning for Epilepsy and Neurodegeneration

Epilepsy continues to be one of the most researched EEG disorders and is frequently used to benchmark performance enhanced with interpretability [7,8]. Byeon et al. EEG-informed stimulation modeling for epileptic disorder detection [14]; Shawly and Alsheikhy [42] propose an attention-based seizure detection model with interpretability via direct SHAP. Meanwhile, Waqar Khan et al. Furthermore, we still need transparent differential diagnosis of AD and frontotemporal dementia (FTD) rather than binary screening, as demonstrated by [43], who propose an explainable and efficient EEG framework for the above-mentioned disorders. Alturki et al. Wavelet-based decomposition combined with intelligent classifiers still performs competitively, as shown directly by [17] and indirectly by Uyanik et al. Epilepsy, depression, and AD are at the forefront of the literature [18], but many other conditions have not been studied at all. These works together motivate the need for explainable temporal modelling (e.g., attention-based sequence learners) and robust cross-dataset validation as fundamental design principles for the next-generation EEG diagnostic AI.

2.8 Explainability in Neuroimaging and Multimodal Brain Disorder Modelling

While QuantumNeuroXAI is limited to brain signals (EEG-centric), research on neuroimaging XAI offers transferable lessons in multimodal fusion, explanation faithfulness, and clinical usability. Odusami et al. Zeineldin et al. [30] [40] show explainable fusion of PET and MRI for AD. NeuroXAI-like explanation toolkits for MRI classification/segmentation workflows based on [41]: Gaur et al. [46], Zeineldin et al. By [41] and Li and Dib [51], all argue that explainability should not be limited to an illustrative remote output but rather be part of the clinical workflow (error triage, confidence calibration, and model debugging). Rahman et al., for example, are particularly proactive about brain tumour XAI. [35], Muhammad et al. [37], Zeineldin et al. [41], Nahiduzzaman et al. [45], and Shuvo et al. Interpretability or Trust‖SHAP justification for predictions [55]. Radiology-focused syntheses importantly caution that within-node acceptance of explanations will depend on clinicians' expectations and domain norms (Hafeez et al. [49]; Rodríguez Mallma et al. [60]) —a pertinent lesson for EEG, for which neurologists may favour channel-level or frequency-band attributions over abstract latent-space explanations. Shon et al. For instance, [53] proposed an explainable normative modelling framework to learn an embedding of resting-state fMRI data for brain disorder prediction. They directly model atypical brain connectivity patterns and can interpret each subgraph for region-level and connection-level explanations. Our work emphasises the use of explainable, unsupervised, and semi-supervised learning for identifying neurological disorders, and how transparency can be ingrained into model design rather than added post hoc.

2.9 Multimodal, Non-EEG Biomarkers and Explainability Extensions

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Recent disorder detection focuses on the use of non-EEG modalities and multimodal biomarkers and is often accompanied by explainability to build trust. Dentamaro et al. Multimodal deep learning combined with XAI is used for early detection of Parkinson's in [47], and voice explainable frameworks for neurological screening are presented by Vizza et al. [38] and Mortezaagha et al. [58]. Coluzzi et al. reported all MRI measures of XAI associated with Alzheimer's biomarker estimation [57], and Khosroshahi et al. We do recognise that the review article [54] explains explainable AI in AD neuroimaging. Similarly, models also exist in the multimodal clinical + imaging framework with explainability (Al-bakri et al. [59]), indicating an aligned path forward: future diagnostics that combine heterogeneous biomarkers with transparent evidence synthesis, rather than isolated classifiers. These works inspire caution in the interface design of EEG-centred systems (e.g., QuantumNeuroXAI) regarding how explanations are delivered, multimodal evidence is integrated, and uncertainty is communicated.

2.10 Key Gaps and Motivation for QuantumNeuroXAI

There are four recurring gaps across the reviewed literature that motivate QuantumNeuroXAI. This includes an absence of reproducible validation (often without cross-subject robustness) of the strong results reported by many EEG deep models limiting their clinical viability,[23]; Rivera et al. [24]; Uyanik et al. [18]). Disease-to-disease discrimination and multi-class settings are also of clinical interest, but they have not been sufficiently addressed (Altıntop [5]; Ahmed et al. [22]). Third, explainability is often an afterthought and is seldom measured for stability, faithfulness, and clinical usability; recent reviews call for standardised XAI evaluation protocols (Presacan et al.). [32]; Bibi et al. [31]; Mehmood et al. [34]). Fourth, quantum-inspired EEG learning is appealing yet limited in its breadth; it requires systematic integration with current deep temporal learning and explainability to provide a foundation for patient-specific discrimination (Saranya and Menaka [4]; Belkacem et al. [16]; Chai et al. [15]).

Table 1: Summary of Prior EEG/Neuroimaging AI Studies and Identified Research Gaps Motivating QuantumNeuroXAI

Ref	**Data / Modality**	**Core Method**	**Key Contribution**	**Research Gap**	**How It Motivates QuantumNeuroXAI**
Alturki et al. [10]	EEG	DWT + Bandpower + SVM	Strong classical EEG diagnostic baseline with engineered time–frequency features	Handcrafted features; limited scalability to cross-subject/general settings; weak end-to-end learning	Justifies hybrid pipelines and robust baselines for ablation vs quantum–DL models
Oh et al. [26]	EEG	Deep CNN on EEG-derived representations	Demonstrates that end-to-end DL can outperform traditional ML for disorder detection	Black-box behaviour; limited clinical transparency; sensitivity to noise/domain shift	Necessitates built-in explainability and robust representation learning
Alves et al.	EEG	Functional connectivity	Uses connectivity-	Connectivity construction	Supports adding structured

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[21]		matrices + CNN	aware learning to capture network-level brain organisation	choices affect results; interpretability remains limited.	(connectivity/graph) priors + explanation at the region/connection level
Saranya & Menaka [4]	EEG	QSVM / QNN with quantum feature mapping	Shows quantum-inspired classifiers can improve separability with compact EEG channels	Narrow task scope; limited interpretability and deployment guidance	Motivates quantum-enhanced representations with explicit XAI for trustworthiness
Mishra & Das [56]	EEG	Explainable epilepsy detection (VAE/Sparse AE + ML)	Integrates explainability (e.g., SHAP-style insight) with strong epilepsy detection	Often, post-hoc explainability, limited generalisation across disorders	Encourages multi-level explainability integrated into model evaluation and reporting
Waqar Khan et al. [43]	EEG	Explainable, efficient DL for AD/FTD	Balances efficiency and explainability for neurodegenerative diagnosis	Disorder-specific; limited multi-disorder extensibility and unified modelling	Supports designing a unified framework adaptable across multiple neurological conditions
Hashem et al. [50]	EEG (BCI tasks)	Explainable ML-based BCI classification	Provides interpretable EEG modelling for motor-task classification in BCI settings	Focused on task decoding, not disorder diagnosis; scalability unclear	Reinforces clinician-facing explanation needs for actionable neuro-AI interfaces
Shon et al. [53]	rs-fMRI	Explainable normative modelling + subgraph explanations	Models deviations from “normality” with region/connection-level explanations	Mostly neuroimaging; EEG integration not addressed; complexity of normative baselines	Inspires intrinsic explainability (not just post-hoc) and deviation-based reasoning in brain disorder detection

In Table 1, we compare and contrast notable EEG and neuroimaging AI studies across modalities, core methods, contributions, and key limitations. Formalising the identified gaps justifies the design of QuantumNeuroXAI: a quantum-enhanced, explainable, and generalizable system for detecting brain disorders. Here, QuantumNeuroXAI is framed as a comprehensive quantum–deep learning pipeline integrating (i) noise-aware EEG preprocessing and representation learning, (ii) quantum-inspired, feature/kernel transformations to achieve separability in high-dimensionality spaces, and (iii) multi-level (channel/frequency relevance and sample-specific rationales) explainability to enable trustworthy detection of neurological disorders.

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3. PRELIMINARIES

This section presents the theoretical and conceptual foundations for understanding the proposed QuantumNeuroXAI framework. It describes the representation of EEG signals across multiple channels, provides an abstraction of the brain signal analysis problem, and reviews central ideas of quantum-inspired machine learning and explainable AI that form the foundation of the proposed approach.

3.1 EEG Signal Representation and Problem Formulation

Electroencephalogram (EEG) signals are measured from multichannel time-series data, which represent the electrical activity of the brain recorded at spatially distributed electrodes. We represent an EEG recording as in Eq. (1).

$$X = \{x_c(t)\}_{c=1}^{C}, \qquad t \in [0,T], \tag{1}$$

where C is the number of EEG channels, $x_c(t)$ is the signal from channel c and is the duration of the record. We apply a sliding or non-overlapping windowing strategy to ensure that inputs to datasets are the same size and to allow learning under non-stationary dynamics (even when the environment changes over time), segmenting each continuous recording into fixed-length segments. Any segmented EEG instance is represented as $X^{(i)} \in R^{C\times L}$, where L is the number of time samples per segment and i is the segment index.

To extract temporal and spectral information from brain activity, a time–frequency representation is computed for each EEG segment $X^{(i)}$. Spectral operators $T(\cdot)$, such as the short-time Fourier transform or wavelet transform, map the signal into a joint time-frequency space, yielding a tensor representation of a segment as in Eq. (2).

$$Z^{(i)} = T(X^{(i)}) \in R^{C\times F\times \tau}, \tag{2}$$

Denotes the frequency bins and represents temporal frames. While this formulation maintains the organisation of spatial channels, it enhances the signal with frequency domain information that is important for differentiating between pathological and cognitive brain states.

This work involves a learning task that concerns the detection of neurological disorders as well as cognitive state classification. For a given EEG-derived representation $Z^{(i)}$, the task is to predict a corresponding label $y^{(i)} \in Y$, with, $Y = \{0,1\}$ for the case of binary disorder detection, or $Y = \{1,\ldots,K\}$ for multi-class cognitive or motor imagery classification. Therefore, this problem can be formally defined as learning a discriminative mapping as expressed in Eq. (3).

$$f_\theta : Z^{(i)} \rightarrow y^{(i)}, \tag{3}$$

where $f_\theta(\cdot)$ stands for the proposed hybrid quantum–deep learning model parameterized as θ

Given a training dataset, the learning task is to find the optimal model parameters by minimizing an empirical risk. Let $D = \{(Z^{(i)},y^{(i)})\}_{i=1}^{N}$ be the set of labeled EEG segments. The optimization problem is formulated as in Eq. (4).

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$$\theta^* = \underset{\theta}{\operatorname{argmin}} \frac{1}{N} \sum_{i=1}^{N} L(f_\theta(Z^{(i)}), y^{(i)}), \tag{4}$$

where $L(\cdot)$ is a loss on this task (e.g., binary cross-entropy, categorical cross-entropy, and so on, depending on the classification setting).

To make things clearer, we summarize the main notations used in the section as follows: X is raw multi-channel EEG signals, as defined in (1); $X^{(i)}$ is temporally segmented EEG windows; $Z^{(i)}$ is the corresponding time–frequency tensor gained via transformation in (2); $y^{(i)}$ is the corresponding class label; and f_θ is the predictive model, defined in (3) and optimized according to (4). These equations formulate the mathematical basis for the QuantumNeuroXAI methodology presented.

3.2 Quantum-Inspired Learning and Explainable AI Foundations

Classical learning model relies on sufficient training samples and linear combination of features; however, high dimensional EEG representations are filled with strong nonlinearity, inter-channel dependency, and dramatically limited labeled samples, due to multi-channel and time–frequency transformation [12]. This leads to appreciable burden of motivation on the types of quantum-inspired learning that we carry forward as being able to have a principled push to mitigate these challenges drawing on concepts like superposition and high-dimensional embedding to help with the added representational capacity. Quantum-inspired models allow for more expressive discrimination of subtle neural patterns typically difficult to disentangle using traditional feature mappings by projecting classical EEG features into an abstract high-dimensional space.

Let (dot) $z^{(i)} \in R^d$ be a vectorized or pooled feature vector of an EEG tensor (representation defined in (2)). That is, quantum-inspired feature encoding gets mapped $z^{(i)}$ into a higher-dimensional Hilbert space H through a nonlinear transformation $\Phi(\cdot)$ as in Eq. (5).

$$\psi^{(i)} = \Phi(z^{(i)}) \in H, \tag{5}$$

with $\psi^{(i)}$ being the quantum-inspired structured state. This mapping mimics the quantum state preparation layout where the amplitude and phase information of classical features embedded within the state is transformed into the normalized shape of embedding which allows for implicit characterization of complex registered correlations across either EEG channels or frequency components. The inner products in H correspond to more expressive similarity measures than those in the original feature space.

The presented framework is based on a hybrid quantum–classical learning paradigm in which quantum-inspired representations are not used on their own, but in conjunction with classical deep neural networks. In particular, we apply a measurement or projection operator $M(\cdot)$ to the encoded representation $\psi^{(i)}$ that results in a real-valued feature vector for downstream learning as in Eq. (6).

$$\tilde{z}^{(i)} = M\left(\psi^{(i)}\right), \tag{6}$$

Afterwards, classical neural layers process it. With this hybrid implementation, the model can gain from the expressivity of quantum-inspired embeddings while maintaining the scalability and optimization stability of classical deep learning models.

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In the classical learning component, attention mechanism is utilized to better predict the temporal dependencies behind the EEG signals. In the case of a sequence of latent representations $\{h_t\}_{t=1}^{\tau}$ obtained from temporal convolutional or recurrent layers, an attention mechanism applies adaptive importance weights across time steps, enabling the model to concentrate on temporally salient regions that are clinically informative. We calculate the attention-weighted representation as in Eq. (7).

$$h^* = \sum_{t=1}^{\tau} \alpha_t h_t, \sum_{t=1}^{\tau} \alpha_t = 1, \tag{7}$$

where α_t indicates the attention weight at time step t. It also introduces temporal interpretability and robustness on EEG non-stationarity as an algorithm enhancing classification performance.

Explainable AI is important, especially in clinical tasks, to provide transparency and trust for neural decision-making. The objective of XAI techniques is to represent in form of an importance score the contribution of input features, intermediates representations, or model components to a particular prediction. For a trained model f_θ, explainability methods search for an attribution function $E(\cdot)$ that returns an explanation vector as in Eq. (8).

$$e^{(i)} = E(f_\theta, z^{(i)}), \tag{8}$$

$e^{(i)}$ in which emphasizes the specific importance of signal dimensions, temporal segments or learned features that drive model output They give information about model behavior outside accuracy metrics.

At least within the field of clinical brain signal analysis, trust cannot be adequately established with a single level of explanation. Thus, we approach this work adopt a multi-level explainability perspective including signal-level, model-level and quantum-representation-level explanations. Signal-level explanations connect predictions to EEG channels, time windows and frequency bands; model-level explanations identify internal model attention and feature attribution activity; and quantum-level explanations evaluate the sensibility of user-encoded dimensions in the quantum-inspired weight space. These complementary perspectives support clinically meaningful interpretation and strengthen the rationale for automated neurological disorder detection systems. In Table 2 the mathematical notations for EEG modeling and the three functions for quantum encoding, hybrid learning and explainability at multiple levels in QuantumNeuroXAI are summarized.

Table 2: Notations Used in the QuantumNeuroXAI Framework

Notation	**Meaning**
$X = \{x_c(t)\}_{c=1}^{C}, t \in [0,T]$	Raw multi-channel EEG; C: channels, T: duration
$A(\cdot), \hat{X}$	Artifact suppression operator; normalized EEG signal
$L, S, X^{(i)}$	Window length, stride, and i-th segmented EEG window
$T(\cdot), Z^{(i)} \in R^{C\times F\times\tau}$	Time–frequency transform and tensor; F: frequency bins, τ: frames
$P(\cdot), U^{(i)}$	Projection/standardization operator and final model input tensor
$\Phi(\cdot), \psi^{(i)} \in H$	Quantum-inspired encoding and encoded state in Hilbert space H

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$Q_\theta(\cdot), M(\cdot), \tilde{z}^{(i)}$	Quantum feature map, measurement/projection, and measured quantum feature vector
$f_\theta(\cdot), \theta$	Hybrid quantum–deep model and trainable parameters
$y^{(i)} \in Y, \hat{y}^{(i)}$	True label and predicted label; Y is class set
$L(\cdot), \eta$	Loss function and learning rate
$\alpha_t, f^{(i)}$	Attention weight at time step t and fused quantum–classical feature
$R^{(i)}_{sig}, R^{(i)}_{mod}, R^{(i)}_{q}, R^{(i)}$	Signal-, model-, quantum-level explanations and unified explanation

4. MATERIALS AND METHODS

This section describes the materials and methods of the proposed QuantumNeuroXAI framework. It outlines available public datasets, preprocessing and feature construction methods, quantum-inspired encoding mechanisms, a hybrid quantum–deep learning architecture, explainability design, and experimental protocols used to ensure robust, reproducible, and interpretable analysis of brain signals.

4.1 Overview of the Proposed QuantumNeuroXAI Framework

The QuantumNeuroXAI framework proposed in this paper is a modular end-to-end system capable of converting raw EEG recordings into clinically meaningful predictions via a quantum–deep learning and explainability pipeline. It takes multi-channel EEG signals from heterogeneous datasets as input. It passes them through a series of stages that are tightly coupled to maintain consistent processing of signal variability, high-dimensional feature extraction, and interpretable classification.

Raw EEGs are fed into the pipeline, and the preprocessing operations — temporal segmentations, normalisations, and time–frequency transforms — serve as the first standardisation steps. These procedures translate raw, continuous EEG data into coherent representations that preserve temporal dynamics, spectral features, and channel spatial relationships. These representations act as a unified intermediary between classical known signal representations and unknown, learning-based modelling structures, enabling analysis that can be conducted irrespective of the underlying dataset.

Once the data is preprocessed, the framework performs quantum-inspired feature encoding that utilises the quantum concepts of superposition and entanglement to map classical features derived from EEG into a sufficiently abstract, higher-dimensional representational space. The expressiveness of EEG features is improved at this stage by modelling complex nonlinear dependencies among channels and frequency bands. These quantum-encoded representations are then interfaced to a hybrid quantum–deep learning core, in which temporal convolutional and attention-based recurrent layers learn short-term fluctuations and long-range neurological patterns in tandem. The hybrid structure enables the framework to leverage the representational richness of quantum-inspired circuits and the optimisation power of classical deep neural networks.

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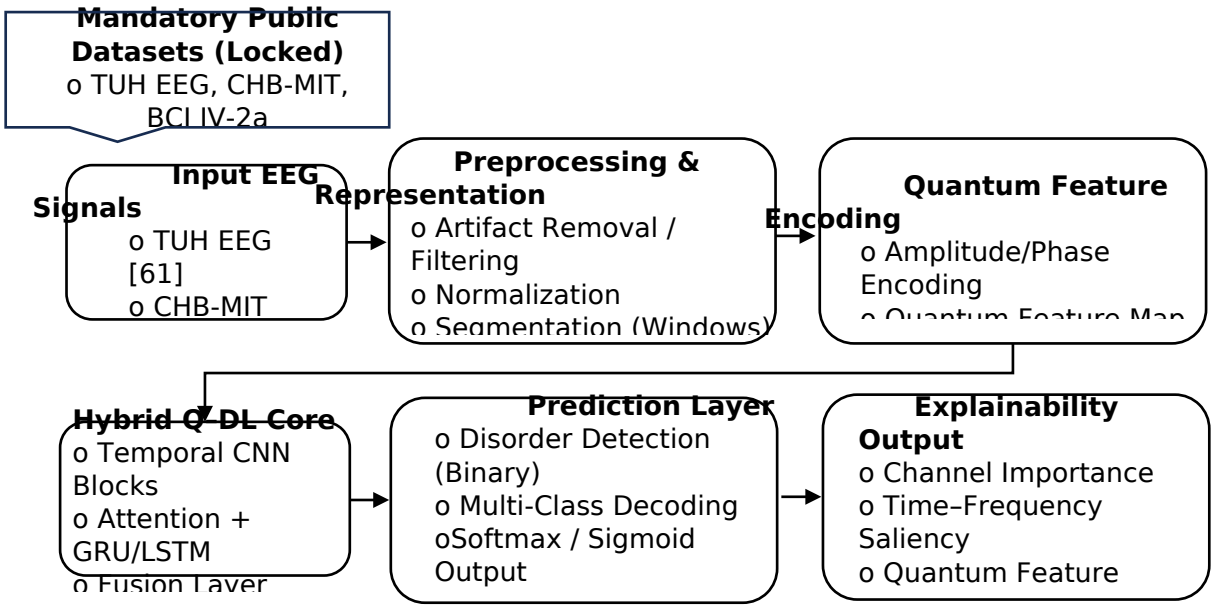

Figure 1: System Architecture of QuantumNeuroXAI for Explainable EEG-Based Neurological Disorder Detection Using Hybrid Quantum–Deep Learning

In the framework, prediction and interpretation are regarded as one inseparable outcome. In addition to classification outputs, such as neurologic disorder detection or cognitive state recognition, QuantumNeuroXAI also embeds Explainable Artificial Intelligence (XAI) mechanisms that provide interpretable insights at multiple levels. These include signal-level, model-level, and quantum-representation-level explanations produced in parallel that each contextualise predictions in terms of EEG channels, temporal segments, and frequency components (signal-level), or feature encodings (model-level and quantum-representation-level). Figure 1 depicts the overall system architecture and the data flow model between these modules.

QuantumNeuroXAI has a single, unified design driven by the desire to be both methodologically rigorous and extensible. This modularity in configuration is available whilst keeping them end-to-end connected, and enables future scaling with alternate quantum encodings, different deep learning backbones or additional explainability techniques without the need to overhaul the system. It enables scalable experimentation, cross-dataset evaluation, and simple deployment in clinical brain signal analysis scenarios via its modular, integrated architecture.

4.2 Dataset Details

We used three publicly available EEG datasets to ensure that the experiments can be fully reproduced and compared between researchers in the accessible data, time, and code.) Together, they encompass clinical variability at scale, seizure manifestations specific to each disorder, and a standard cognitive decoding benchmark. TUH EEG offers a wide variety of clinical EEG recordings that are ideal for robust representation learning and scalability analysis [61]. For example, CHB-MIT provides expert-annotated seizure events for the detection of neurological disorders and segment-level evaluation [62]. The BCI Competition IV Dataset 2a uses a widely adopted multi-class motor imagery protocol [63] and enables validation of general brain-signal decoding performance. The study is limited to public datasets, thereby eliminating unverifiable private cohorts and areas of reporting ambiguity, and promoting transparency by enabling preprocessing, training, and explainability analysis to be replicated in a controlled environment. Dataset characteristics, task alignment, and the official download source used in QuantumNeuroXAI are summarised in Table 3.

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Table 3: EEG Datasets, Task Alignment, and Official Download Sources Used in QuantumNeuroXAI

Sl. No	Dataset	Signal Type	Primary Task Alignment	Labels / Annotation Type	High-Level Scale	Download URL	Role in Quantum NeuroXAI
1	TUH EEG (Temple University Hospital EEG Corpus) [61]	Clinical scalp EEG	General clinical EEG modeling; robustness under heterogeneity	Clinical metadata and heterogeneous recording contexts (varies across subsets)	Large-scale clinical corpus	https://isip.piconepress.com/projects/tuh_eeg/	Pretraining /robust feature learning; stress-testing generalization under real-world variability
2	CHB-MIT Scalp EEG Database [62]	Scalp EEG	Neurological disorder detection (epileptic seizure detection)	Seizure onset/offset annotations	Moderate-scale, subject-level recordings	https://physionet.org/content/chbmit/1.0.0/	Disorder-specific supervised learning; primary benchmark for seizure detection and explainability case studies
3	BCI Competition IV Dataset 2a [63]	EEG (+EOG)	Cognitive state decoding (motor imagery classification)	Multi-class motor imagery labels	Standard benchmark cohort	https://www.bbci.de/competition/iv/#download	Cross-task validation; evaluates whether quantum encoding and XAI remain consistent beyond pathology detection

The experimental design allows for an extensive yet controlled evaluation of QuantumNeuroXAI in terms of clinical realism, disorder detection fidelity and general EEG decoding capability, all while being fully reproducible due to the exclusive use of public data61–63 HB (2012).

4.3 EEG Preprocessing and Feature Construction

Unprocessed EEG recordings undergo an initial step to suppress and normalize artifacts to minimize the external noise introduced by physiological and environmental sources. Define the raw multi-channel EEG signal as (1) A preprocessing operator $A(\cdot)$ is then

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used to perform artifact suppression by diminishing high-amplitude transients and low frequency drifts, resulting in a denoised signal as in Eq. (9).

$$X' = A(X). \tag{9}$$

then, to mitigate inter-subject and inter-session variability, channel-wise normalization is performed. Hence, each channel signal $x_c'(t)$ is normalized with respect to its empirical mean μ_c and standard deviation σ_c, as expressed in Eq. (10).

$$\hat{x}_c(t) = \frac{x_c'(t) - \mu_c}{\sigma_c}, \tag{10}$$

where $\hat{x}_c(t)$ is the normalized EEG signal for channel.

Following this process, the normalized EEG signals are segmented into fixed-length temporal windows, providing local stationarity of the EEG signals, while also allowing learning of their patterns in a batch-wise manner. For a window length L and stride S, the matrix representation of segmented EEG is defined as in Eq. (11).

$$\hat{X}^{(i)} = \{\hat{x}_c(t)\}_{c=1}^{C}, \qquad t \in [iS, iS + L], \tag{11}$$

where i indexes the temporal segment. This segmentation strategy affords significant freedom in both controlling temporal resolution, and ensuring that enough contextual information is retained in each segment to construct identifiable patterns for neurological classification.

In order to provide spectral features for the temporal EEG segments, a time–frequency transformation is performed for every segmented time window. The mapping results in a joint time–frequency representation using the spectral operator $T(\cdot)$ defined in Eq. (12).

$$Z^{(i)} = T(\hat{X}^{(i)}), \tag{12}$$

where $Z^{(i)} \in R^{C \times F \times \tau}$ encodes activity in the frequency domain, across F frequency bins, and τ temporal frames. This allows the model to learn discriminative oscillatory patterns related to neurological disorders and cognitive states.

The obtained time–frequency representations are then structured into tensor blocks on a channel-wise basis to enable deep learning inputs. The tensor has preserved the spatial ordering of EEG channels while keeping temporal and spectral continuity for each segment i. This tensor $Z^{(i)}$ can then optionally reshaped or projected (through a deterministic mapping $(\cdot)$) into a compact feature tensor $U^{(i)}$ that is, when the learning architecture demands such transformation as expressed in Eq. (13).

$$U^{(i)} = P(Z^{(i)}), \tag{13}$$

where $U^{(i)}$ is the uniform format of the data used in the following quantum-inspired encoding and hybrid learning phases. The pipeline preserves required temporal, spectral and spatial features of the EEG signals for learning and explainability subsequently providing a solution that maintains consistency across datasets.

4.4 Quantum-Inspired Feature Encoding Module

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After preprocessing EEG and constructing the features, to increase the representational ability of classical EEG features, the proposed framework adapts a quantum-inspired feature encoding module. Define $U^{(i)} \in R^d$ the vectorized or pooled feature representation from (13) as The classical to quantum mapping is intended to embed $U^{(i)}$ into an higher dimensional abstract space, which reproduces properties for the representation of quantum states. It is this mapping (which we will describe mathematically by a nonlinear embedding function $\Phi(\cdot)$), which produces a quantum-inspired state as in Eq. (14).

$$\psi^{(i)} = \Phi(U^{(i)}), \tag{14}$$

$\psi^{(i)} \in H$ and H represents a Hilbert space with higher dimensionality than the original feature space.

Amplitude and phase encoding patterns are what we use to build $\psi^{(i)}$. In amplitude encoding, we map normalized feature values to the magnitudes of the resulting state components to make sure the whole representation respects a unit-norm constraint. For a normalized feature vector $\tilde{U}^{(i)}$, the amplitude-encoded state is given by Eq. (15).

$$\psi_{amp}^{(i)} = \sum_{k=1}^{d} \tilde{u}_k^{(i)} \, |k\rangle, \tag{15}$$

where $|k\rangle$ represents a unit vector in H. Phase encoding adds phase offsets to each Fourier-transformed feature to preserve relative relationships among features, resulting in a state that is a composite of magnitude and relational information among the Fourier-transformed EEG derived feature set.

Afterwards, the encoded state is iteratively transformed through a parameterised quantum feature map, which simulates the quantum circuit evolution. This transformation is modeled as in Eq. (16).

$$\phi^{(i)} = Q_\theta\left(\psi^{(i)}\right), \tag{16}$$

where $Q_\theta(\cdot)$ is a parameterized mapping and controls the complexity of feature interactions Projecting the encoded state into a more expressive representational subspace implicitly models higher-order correlations among EEG channels, frequency components, and temporal features in this step.

We obtain real-valued features using a measurement (or a projection) operation on the classical domain since downstream learning components work on real-valued features. This measurement simulation computes expectation like values from the transformed state, yielding the expression in Eq. (17).

$$\tilde{z}^{(i)} = M\left(\phi^{(i)}\right), \tag{17}$$

with where $\tilde{z}^{(i)} \in R^{d'}$ is a compact representation with real values, which is suitable to survive classical deep neural processing. The measurement operator $M(\cdot)$ has an aim to keep discriminative information without reducing the dimensionality and optimising the dimensions deterministically (invariance).

For those computational considerations, the quantum inspired encoding module is intended to maximize expressiveness without sacrificing efficiency. The intermediate representation exists in a high-dimensional Hilbert space, but a measurement collapses that representation into a compact feature vector that is well-suited for large-scale training. The framework does not require near-term quantum devices as it simulates

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quantum feature maps on classical hardware, taking advantage of quantum-inspired nonlinear embeddings. This architecture allows efficient incorporation of quantum-inspired representations into standard deep learning pipelines with minimal computational overhead.

The proposed quantum-inspired encoding is implemented entirely on classical hardware through deterministic simulations of quantum feature mappings. This design ensures that the framework remains computationally feasible and deployable in real-world clinical environments without reliance on quantum computing infrastructure.

4.5 Hybrid Quantum–Deep Learning Architecture

The hybrid quantum–deep learning architecture represents the central predictive element of QuantumNeuroXAI and is structured to jointly leverage quantum-inspired feature representational power and classical deep neural modeling capabilities. We can use the notation in (17) where $\tilde{z}^{(i)} \in R^{d'}$ is the quantum-inspired feature vector that we obtain from the encoding and measurement process. This embedding is subsequently reshaped into a format suited for deep temporal modeling, where they are fed into the hybrid architecture seen in Figure 2.

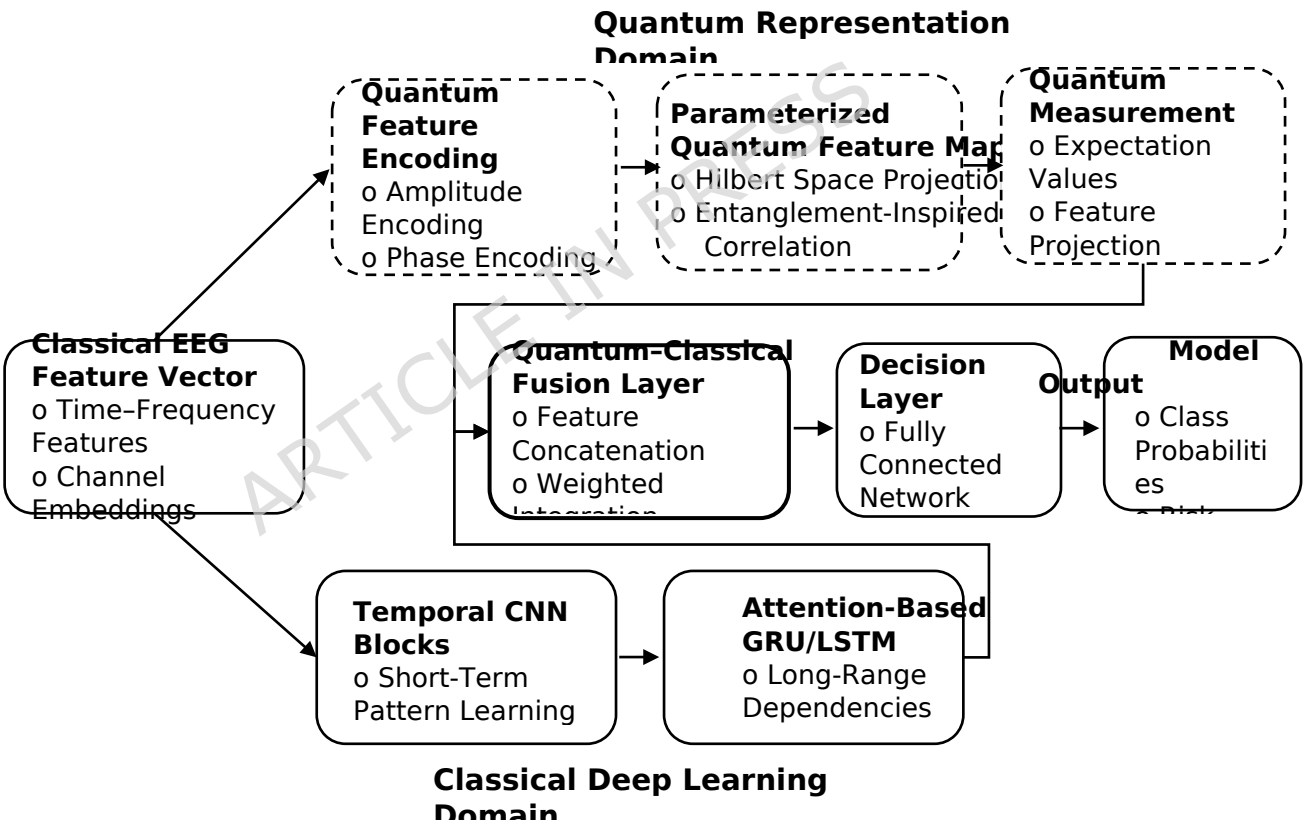

Figure 2: Hybrid Quantum–Deep Learning Fusion Architecture for Temporal EEG Representation Learning and Robust Neurological Disorder Classification

We utilize the so-called temporal convolutional learning blocks as the first stage of a classical deep processing, capturing short-term temporal dependencies and local patterns in the EEG dynamics. Temporal convolution operation with kernel W_k given the input sequence $\{\tilde{z}_t^{(i)}\}_{t=1}^{\tau}$ to produces intermediate feature maps as in Eq. (18).

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$$h_t^{(k)} = \sigma\left(\sum_{j=0}^{K-1} W_k(j)\, \tilde{z}_{t-j}^{(i)} + b_k\right), \tag{18}$$

where K is the size of the kernel, b_k is a bias, and $\sigma(\cdot)$ is an activation non linearfunction. By stacking several temporal convolutional layers, temporal features of ever more abstract nature can be learned hierarchically from the quantum-inspired input sequence.

The architecture combines attention-based recurrent modeling in layers that capture long-range temporal dependencies and contextual interactions in EEG segments. The convolved feature sequence $\{h_t\}$ is fed to a recurrent unit (e.g. a gated recurrent unit or long short term memory network), resulting in hidden states $\{s_t\}$. An attention mechanism then assigns adaptive importance weights to these hidden states, thus enabling the model to focus on temporally salient regions. This gives us the resulting context vector computation as in Eq. (19).

$$s^* = \sum_{t=1}^{\tau} \alpha_t s_t, \tag{19}$$

where, α_t is normalized attention weight for time step t. It better predictive ability and interpretation power of prediction result, due to explicitly capturing the temporal dependence in predictions.

The core novelty of the architecture is a novel quantum–classical hybrid fusion that combines information from quantum-inspired encoding with classical deep temporal modeling. $\grave{z}^{(i)}$ denote a pooled representation obtained directly from the quantum-inspired features and (19) s^* denotes the attention-enhanced temporal representation. A concatenation and transformation operation that fuses these two representations is defined as in Eq. (20).

$$f^{(i)} = \varphi\left([\grave{z}^{(i)} \,\|\, s^*]\right), \tag{20}$$

where $[. \| .]$ means vector concatenation and $\varphi(\cdot)$ is a trainable transformation that balances the contributions of quantum-inspired and classical features. This allows the model to leverage both high-dimensional nonlinear embeddings and temporally contextualized representations together.

The fused representation $f^{(i)}$ is next fed to the decision and output layers, which perform task-specific classification. The last prediction is simply done by a fully connected layer then softmax or sigmoid as in Eq. (21).

$$\hat{y}^{(i)} = g\left(W_o f^{(i)} + b_o\right), \tag{21}$$

where $g(\cdot)$ represents the output activation function, and $W_o$and b_o the learnable parameters. This formulation enables binary detection of neurological disorder and multi-class classification of cognitive state.

Interactions of temporal convolution, attention-based recurrence, quantum-inspired encoding, and fusion layers are visually summarized in Figure 2. The architecture guarantees that quantum-inspired representations are not standalone entities but play an active role alongside classical deep learning modules, leading to a unified hybrid model that balances expressiveness, interpretability and computational tractability.

4.6 Explainability and Interpretability Design

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So, we have added explainability as an important component to the basic design of QuantumNeuroXAI to make sure that model decisions are transparent and trustworthy with clinical relevance. Our framework applies a multi-level interpretability approach that elucidates predictions at the signal, model, and quantum-representation layers, allowing for convergent insights regarding predictions of how neurological decisions are generated. Denote $\hat{y}^{(i)}$ the prediction from (21) for an input EEG segment as $U^{(i)}$.

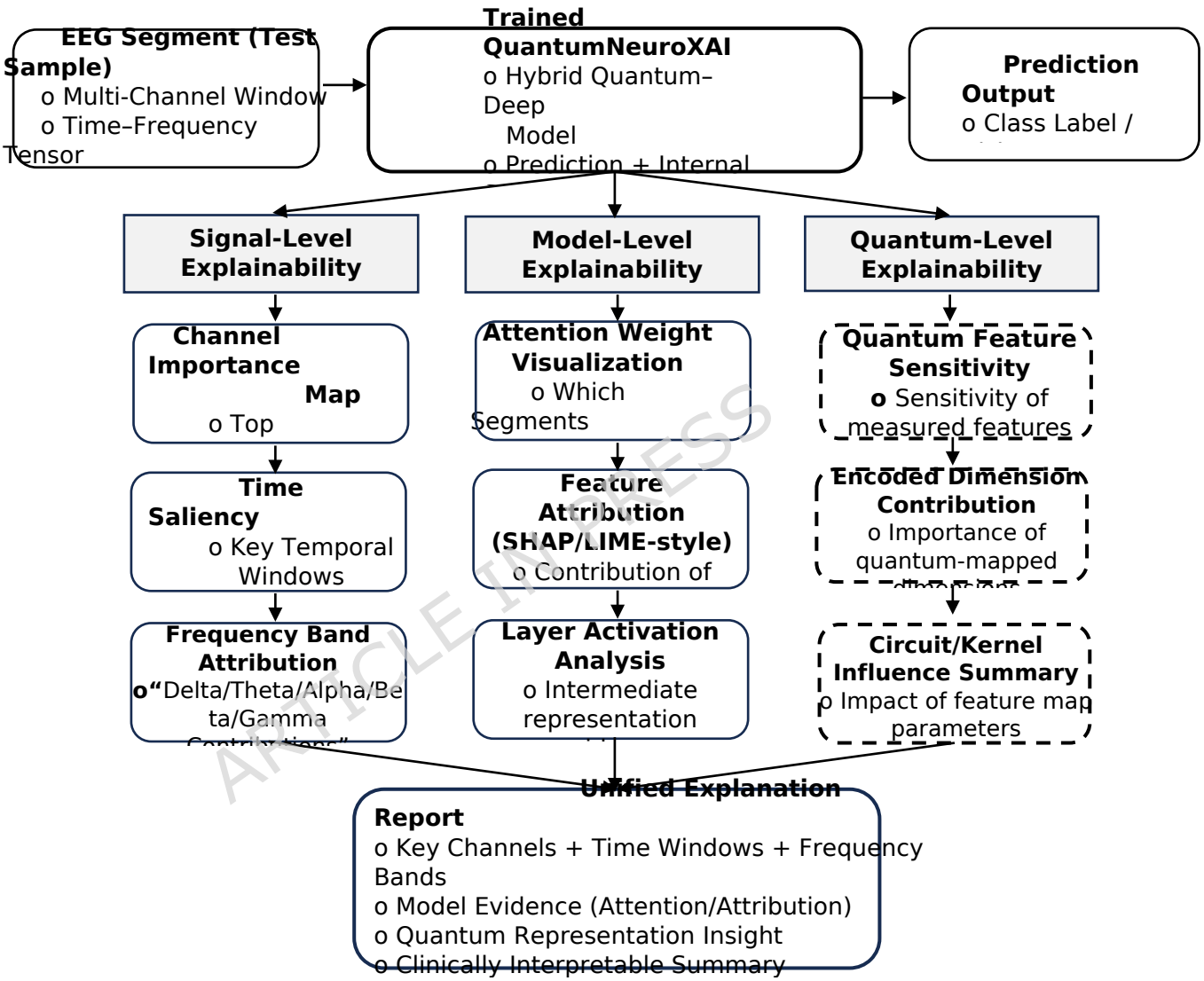

Figure 3: Multi-Level Explainability Workflow of QuantumNeuroXAI for Transparent and Clinically Interpretable EEG-Based Neurological Analysis

Whether it is the raw data or the actual feature space, it has been the pursuit of signal-level explainability since then, i.e., how can we associate physiologically significant EEG descriptors to model predictions. The time–frequency tensor $Z^{(i)}$ defined in (12) is then evaluated with a signal attribution operator $E_{sig}(\cdot)$ to extract relevance scores per channel, temporal window, and frequency band as in Eq. (22).

$$R_{sig}^{(i)} = E_{sig}(f_\theta, Z^{(i)}), \tag{22}$$

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where $R_{sig}^{(i)} \in R^{C \times F \times \tau}$ we use EEG regions that most strongly influence the prediction. Model-level explainability tries to understand what role do internal components of the network play towards the final decision. For example, in particular, attention-based recurrent modeling tends to introduce intrinsic interpretability via attention weights $\{\alpha_t\}$ (19). The weights are used to represent the significance of various parts of time in the decision making. Furthermore, we also use feature attribution methods in intermediate or fused representations $f^{(i)}$ from (20) to measure the contribution from learned features. A vector of model-level explanation is framed as in Eq. (23).

$$R_{mod}^{(i)} = E_{mod}(f_\theta, f^{(i)}), \tag{23}$$

where $R_{mod}^{(i)}$ represents the influence $\hat{y}^{(i)}$ of any internal representation (in this case, attention-driven components) on.

Quantum-inspired representations — that show how models predict — captures classical to quantum-level explainability. An analysis of model sensitivity was conducted over the actual quantum-inspired feature vector $\tilde{z}^{(i)}$ denoted in (17), to assess the impact of changes in encoded dimensions on the prediction. Formally this can be expressed through a quantum attribution operator $E_q(\cdot)$ as expressed in Eq. (24).

$$R_q^{(i)} = E_q(f_\theta, \tilde{z}^{(i)}), \tag{24}$$

$R_q^{(i)}$ describes the relative weight of quantum-inspired encoded features or parameterised feature-map features. As such, this level of explanation sheds light on how downstream decisions are affected by quantum-inspired embeddings, thereby filling an important interpretability gap affording insights with regard to hybrid quantum–classical models.

The framework combines the three explanation levels into a single synthesis workflow, and presents coherent, clinically meaningful interpretations. Example-level relevance information is specific to each EEG segment and is acquired by aggregating the signal-level relevance, the model-level, and the quantum-level relevance info as formulated as in Eq. (25).

$$R^{(i)} = S\left(R_{sig}^{(i)}, R_{mod}^{(i)}, R_q^{(i)}\right), \tag{25}$$

where $S(\cdot)$ is a nonparametric structure aggregation which aligns explanations over the representations and outputs a human-friendly summary. It brings physiological EEG patterns, neural model behavior, and quantum-inspired feature contribution into a cohesive unified explanation.

The full explainability pipeline showing the steps of generating signal-level, model-level, and quantum-level explanations and their combination is shown in Figure 3. QuantumNeuroXAI enables transparent detection of neurological disorders by integrating and simultaneously designing Explainability at multiple representational levels, thus yielding interpretable evidence to accompany automated predictions for ease of interpretation by clinicians and researchers at various levels of expertise.

4.7 Training Strategy and Experimental Protocol

The QuantumNeuroXAI training strategy guarantees fair evaluation, dataset robustness and reproducibility. Define D a labeled EEG dataset consisting of segmented samples as per Eq. (4) as follows. The strata keeping subject level class distributions are split into disjoint training, validation, and test subsets for each dataset. The splits obtained are

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represented as D_{tr}, D_{val}, and D_{te}, ensuring that there is no temporal and subject-level leakage between training and evaluation stages.

The optimization of a model is then done as the minimization of a task-oriented empirical risk over the training set. The hybrid model makes predictions $\hat{y}^{(i)}$ as described in (21) for a batch of samples $(U^{(i)}, y^{(i)}) \in D_{tr}$. We define the objective of optimization as in Eq. (26).

$$J(\theta) = \frac{1}{|D_{tr}|} \sum_{(U^{(i)}, y^{(i)}) \in D_{tr}} L(\hat{y}^{(i)}, y^{(i)}) + \lambda \Omega(\theta), \tag{26}$$

where $L(\cdot)$ is the classification loss, $\Omega(\theta)$ is a regularization term and λ is a hyperparameter that balances the contribution of that term. For tasks predicting neurological disorder we apply binary cross-entropy loss while for multi-class classification of cognitive state we use categorical cross-entropy loss. The update of model parameters is done via gradient-based optimization with early stopping based on performance of on D_{val} the validation set.

In order to evaluate generalization capabilities beyond dataset-specific properties, we apply a cross-dataset evaluation protocol. In this setting, the model is trained on one or more datasets and evaluated on the corresponding held-out dataset without any fine-tuning, allowing the evaluation of model transferability and robustness to distributional shifts. Let D_s and D_t denote a source dataset and a target dataset, respectively. Cross-dataset performance is computed as in Eq. (27).

$$P_{cross} = E(f_{\theta_s}, D_t), \tag{27}$$

where f_{θ_s} denotes: model for which the D_s and $E(\cdot)$ evaluation Especially when there are heterogeneous acquisition conditions on EEG that would lead to variability in resulting quantum-inspired representations, this protocol is essential for asserting stability.

Experimental settings were carefully controlled, allowing for enforced reproducibility and training stability. For the experiments, we always split the data, initialize parameters and the optimization routine using fixed random seeds. Hyperparameters are chosen to maximize performance on a validation set and do not change across datasets, so that there is no implicit tuning bias. To track training stability, convergence analysis of the loss in (26) and variance across runs are used. Combined, these measures guarantee that these reported results relate to stable behavior of the model instead of stochastic artifacts, and allow for responsible assessment of the suggested QuantumNeuroXAI framework.

4.8 Algorithmic Representation

The following section represents algorithmically the entire QuantumNeuroXAI Framework proposed in the study. It through concise algorithms, formalizes the preprocessing, hybrid quantum–deep learning training, and multi-level explainability procedures, providing clarity to the operational flow, reproducibility, and integration of the main methodological components.

Algorithm: EEG Preprocessing and Feature Construction Pipeline
Input: Raw multi-channel EEG signal X, window length L, stride S, spectral operator $T(\cdot)$
Output: Channel-wise feature tensor $U^{(i)}$

1. Apply artifact suppression to X to obtain cleaned signal X' using $A(\cdot)$.
2. For each channel $c = 1,\ldots,C$, normalize X' using channel-wise mean and

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standard deviation to obtain $\hat{X}$.

3. Segment $\hat{X}$ into windows $\{\hat{X}^{(i)}\}$ of length L with stride S.
4. For each segment $\hat{X}^{(i)}$, compute time–frequency representation $Z^{(i)} = T(\hat{X}^{(i)})$.
5. Organize $Z^{(i)}$ into a channel-wise tensor preserving channel, frequency, and temporal dimensions.
6. Apply deterministic projection $P(\cdot)$ to obtain standardized input tensor $U^{(i)}$.
7. Return $U^{(i)}$ for all segments.

Algorithm 1: EEG Preprocessing and Feature Construction Pipeline

This can be summarized as:Algorithm 1: Mapping from raw multi-channel EEG recordings to canonical feature representations for quantum-inspired and deep learning First, the algorithm suppresses various sources of physiological artifact noise (ambient noise during recordings, sensor drift and electronic noise) followed by channel-wise normalization for inter-subject and inter-session variability reduction. Subsequently, the cleaned EEG signals are sliced into fixed-length temporal windows, allowing localized analysis of non-stationary brain dynamics while maintaining fixed input dimensions. Then each of this EEG segment is converted to a time-frequency representation, a process that adds complementary spectral information to the signals, allowing the signal to become more optimal for neurological and cognitive identification. Lastly, the time–frequency representations are sorted into channel-wise tensors in which spatial, temporal, and frequency relationships across electrodes are preserved. Such a pipeline guarantees the consistency of heterogeneous datasets and a reproducible dataset agnostic foundation to the following quantum-inspired encoding and hybrid learning stages. The algorithm improves transparency and enables fair comparison of experiments by specifying each preprocessing step.

Algorithm: Hybrid Quantum–Deep Learning Training Procedure

Input: Preprocessed EEG feature tensors $\{U^{(i)}\}$, labels $\{y^{(i)}\}$, learning rate η, loss function $L(\cdot)$

Output: Trained model parameters θ

1. Initialize hybrid model parameters θ using a fixed random seed.
2. For each training epoch do:
3. Sample a mini-batch $(U^{(i)}, y^{(i)})$ from the training set.
4. Encode $U^{(i)}$ into a quantum-inspired representation $\psi^{(i)}$ using $\Phi(\cdot)$.
5. Apply the quantum feature map and measurement to obtain $\tilde{z}^{(i)}$.
6. Extract short-term temporal features using temporal convolutional layers.
7. Model long-range dependencies using attention-based recurrent layers.
8. Fuse quantum-inspired and deep temporal representations to obtain $f^{(i)}$.
9. Compute predictions $\hat{y}^{(i)}$ using decision and output layers.
10. Evaluate loss $L(\hat{y}^{(i)}, y^{(i)})$.
11. Update model parameters $\theta \leftarrow \theta - \eta \nabla_\theta L$.
12. End for.
13. Return optimized parameters θ.

Algorithm 2: Hybrid Quantum–Deep Learning Training Procedure

The end-to-end training of the proposed hybrid quantum–deep learning model is summarized in Algorithm 2. The algorithm encodes classical features into quantum-inspired representations based on preprocessing EEG feature tensors to improve expressiveness and ensure it can better fit highly complex nonlinear dependencies. The encoded features then pass through temporal convolutional layers to learn short-term EEG patterns, and attention-based recurrent modeling to capture long-range temporal dependencies. An inverse compatible representation fusion approach aggregates quantum inspired representation with deep temporal features to joint exploit advantages of both representational paradigms. The decision layers consume the fused features to output task-specific predictions. Gradient-based learning allows us to

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optimize model parameters by minimizing a proper classification loss, where the updates are performed iteratively over training epochs. The proposed algorithm explicitly reveals the seamless embedding of quantum inspired encoding into conventional deep learning workflows, scalable on classical hardware, whilst preserving the advantages of quantum inspired feature mappings. At its heart, QuantumNeuroXAI proposes a new methodological concept — structured training procedure.

Algorithm 3: Multi-Level Explainability Generation and Synthesis
Input: Trained model f_θ, EEG feature tensor $U^{(i)}$, time–frequency tensor $Z^{(i)}$, quantum-inspired features $\tilde{z}^{(i)}$
Output: Unified explanation $R^{(i)}$

1. Compute model prediction $\hat{y}^{(i)} = f_\theta(U^{(i)})$.
2. Generate signal-level relevance $R^{(i)}_{sig}$ over channels, time, and frequency using $E_{sig}(\cdot)$.
3. Extract model-level explanations $R^{(i)}_{mod}$ via attention weights and feature attribution on $f^{(i)}$.
4. Perform quantum-level sensitivity analysis on $\tilde{z}^{(i)}$ to obtain $R^{(i)}_{q}$.
5. Align and normalize $R^{(i)}_{sig}$, $R^{(i)}_{mod}$, and $R^{(i)}_{q}$.
6. Synthesize a unified explanation $R^{(i)} = S(R^{(i)}_{sig}, R^{(i)}_{mod}, R^{(i)}_{q})$.
7. Return $R^{(i)}$.

Algorithm 3: Multi-Level Explainability Generation and Synthesis

In summary, our explainability strategy that constitutes QuantumNeuroXAI can be formalized as Algorithm 3, where interpretations at multiple representational levels are synthesized and incorporated. In other words, given a trained model and an input EEG instance, the algorithm first generates a prediction and them obtains signal-level explanations that indicate salient EEG channels, temporal segments and frequency bands contributing towards the decision. We then extract model-level explanations by analyzing attention weights and first-order feature attributions learned inside the deep learning architecture to show which internal representations contribute to the output. Simultaneously, sensitivity analysis of quantum-inspired features enables quantum-level explainability by elucidating the types of encoded dimension and feature-map parameters. They are then aligned, normalized, and aggregated to be included in the same explanatory report that connects physiological behaviors of EEG and supports model reasoning through quantum-inspired representations. The algorithm increases explainability, reliability, and clinical relevance by aggregating explanations across levels, which can help address a major issue of the current black-box EEG analysis models.

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4.9 Performance Evaluation Metrics

QuantumNeuroXAI is evaluated by a broad set of metrics for classification accuracy, robustness on datasets, and the reliability of the explainability outputs. Let $\{(y^{(i)}, \hat{y}^{(i)})\}_{i=1}^{N}$ be the set of true labels and the associated model predictions on a test set where $\hat{y}^{(i)}$ is defined in (21).

Standard metrics are used to measure predictive effectiveness for classification performance. Accuracy can be defined as the ratio of correctly classified samples as in Eq. (28).

$$\text{Accuracy} = \frac{1}{N}\sum_{i=1}^{N} I(y^{(i)} = \hat{y}^{(i)}), \tag{28}$$

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where $\mathbb{I}(\cdot)$ is the indicator function. In addition, for the case of class imbalance, we report precision, recall, and F1-score as well. Precision simply represents how many positive predictions were correct, whereas recall returns the proportion of positives correctly captured. The F1-score, which is the harmonic mean of precision and recall, provides a balanced measure of classification performance as in Eq. (29).

$$F1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}. \tag{29}$$

the area under the receiver operating characteristic curve is used to assess discriminative capability across different decision thresholds for binary neurological disorder detection tasks.

Cross-dataset and cross-task evaluation protocols measure robustness and generalization. Based on (27), let P_{in} and P_{cross} the performance measures in- and cross-dataset be denoted by and, respectively. The generalization gap is computed as in Eq. (30).

$$\Delta_{gen} = |P_{in} - P_{cross}|, \tag{30}$$

where a lower value of Δ_{gen} indicates more robust to the distributional shifts. We further analyze different model performances across multiple runs to assess the stability of the model with stochastic optimization and parameter initialization.

Evaluating that explanation outputs are repeatable and comparable across similar input instances, to ensure that explanations are consistent and reliable. Let the explanation maps where (25) give the explanation maps are $R^{(i)}$. Consistency is computed as the similarity of the explanation maps for a same input when the model is trained repeatedly in the same way or the model is perturbed. Denote $R_a^{(i)}$ and $R_b^{(i)}$ two explanation instances for the same input, and let $|x|$ denotes the number of words in x. Explanation consistency is quantified as in Eq. (31).

$$C_{XAI} = \frac{1}{N} \sum_{i=1}^{N} \text{sim}(R_a^{(i)}, R_b^{(i)}), \tag{31}$$

where $\text{sim}(\cdot)$ is a similarity measure — cosine, rank correlation, etc. Then, based on physiological knowledge about previously established EEG patterns and clinically relevant signal components, the reliability of salient regions is further tested.

Taken as a whole, these evaluation metrics represent a multi-dimensional characterization of QuantumNeuroXAI, one that incorporates not only predictive fidelity, but also robustness, generalization, and the trustworthiness of explainability results which are essential when applied to clinical brain sign data.

4.10 Implementation Details

To ensure transparency and reproducibility, we implemented the proposed QuantumNeuroXAI framework using popular open-source software tools. Python is used to implement all the signal processing and model development functionality, with numerical operations, neural network construction and optimization done using standard scientific computation and deep learning libraries. Standard signal processing utilities are used for time-frequency transformations and preprocessing specific to the EEG signal, and the quantum-inspired feature encoding and hybrid model components are implemented as classical simulations embedded in the deep learning pipeline.

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All experiments are performed on a multi-core CPU and a single high-memory GPU workstation to speed up model training and inference time. All the deep learning components were implemented with GPU acceleration but then quantum-inspired feature mapping and explainability computations were run on the same hardware in the absence of any specialized quantum device. Such configuration renders the proposed framework deployable on a standard computing infrastructure without requiring access to quantum hardware.

In order to ensure clear reproducibility and stable training behaviour, where all model parameters are initialized using deterministic initialization schemes, a fixed random seed is also enforced across the whole data splitting, parameter initialization and overall optimization routine. Hyperparameters like learning rates, batch sizes, and regularization coefficients are chosen based on validation performance but kept constant across datasets. All runs log software versions, hardware configurations, and experimental settings, so that reported results can be precisely replicated and fairly compared when subsequent studies are conducted.

The implementation of the proposed QuantumNeuroXAI framework, including preprocessing, model architecture, and experimental setup, is publicly available at: https://github.com/venkateshwarlu-bondu/QuantumNeuroXAI.

To ensure long-term accessibility and reproducibility, a permanent archived version of the code is also available via Zenodo: https://doi.org/10.5281/zenodo.19200650.

5. EXPERIMENTAL RESULTS

This section describes the experiments which were performed to validate the proposed QuantumNeuroXAI framework. We report quantitative performance, robustness, and generalization results on public EEG datasets, a comparison of the proposed method to the state-of-the-art baseline models, and an analysis of the explainability outputs for interpretability, reliability and practical relevance in the context of neurological disorder detection.

5.1 Experimental Setup and Protocol

QuantumNeuroXAI is experimentally evaluated using three public EEG datasets: TUH EEG [61], CHB-MIT [62], and BCI Competition IV Dataset 2a [63]. Datasets are preprocessed with the unified pipeline described in Section 4.3 to ensure segmented and time–frequency features are constructed in a consistent way across heterogeneous recording conditions. In-dataset evaluation is performed using subject-independent stratified splits to prevent leakage, where the recordings are divided into training, validation such that classes are balanced at the subject level when forming the training, validation and testing subsets. Only to perform the early stopping and select the hyperparameters the validation set is used, while the test set is never used until the final reporting. We supplement in-dataset testing with a cross-dataset protocol to measure robustness under distribution shift. In this setting, the model is trained on one dataset (source) and evaluated on a different dataset (target) without fine-tuning to quantify generalization/transferability of quantum-inspired representations and hybrid modeling of the temporal dimension.

Training is performed using mini-batch optimization with a fixed set of hyperparameters kept constant across datasets to ensure fair comparison. The model is optimized using a gradient-based optimizer with an initial learning rate η and batch size B. Early stopping is applied based on validation loss to prevent overfitting, and standard regularization techniques are used to stabilize training. The specific loss for the task is in line with the formulation of the objective in (26) and depends on the task itself: binary cross-entropy for neurological disorder detection tasks and categorical cross-entropy for multi-class

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EEG decoding. Each experiment is repeated over several runs (with different random seeds) to allow for stability analysis and shows mean performance and a measure of variability.

We implemented all the experiments in Python with common deep learning and science computing libraries. Hybrid model optimization can still be sped up with training and inference running on a GPU-enabled workstation, while quantum-inspired feature encoding and measurement can be performed inside the same classical computing environment. All random seeds for data splitting, parameter initialization, and training routines are fixed to ensure reproducibility as well as all log software versions and experiment configurations to replicate the results.

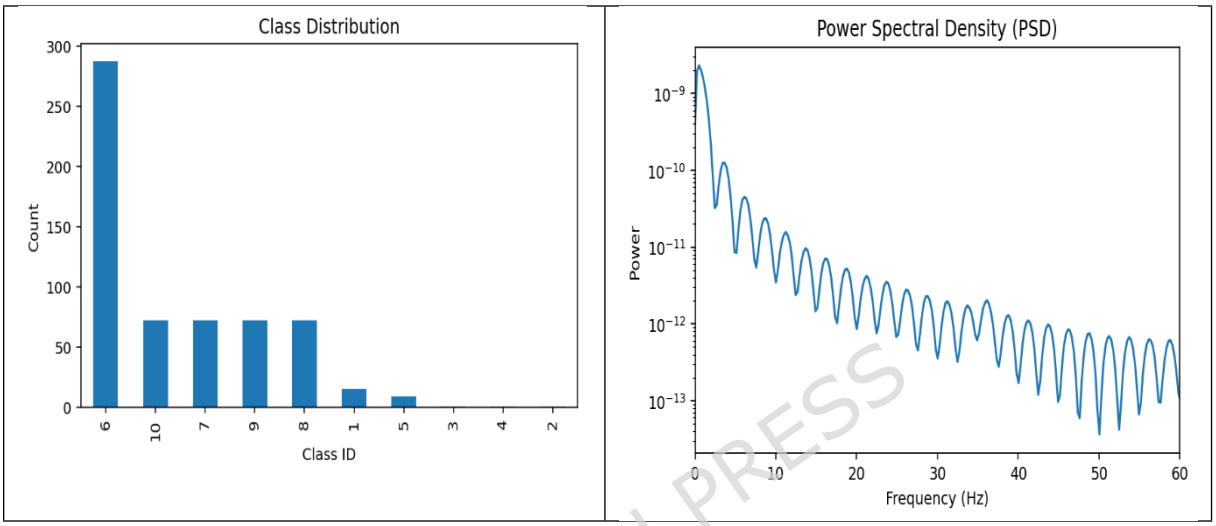

Figure 4: Exploratory Analysis of the BCI Competition IV Dataset 2a: Class Distribution and Spectral Characteristics

An exploratory analysis of the BCI Competition IV Dataset 2a dataset is illustrated in Figure 4.The left panel displays the distribution of C4 and C3 features across the motor imagery classes, showing clear class imbalance. The right panel shows that the EEG spectra show dominant low-frequency components and a characteristic spectral decay (black–blue) that motivate the extraction of time–frequency features in the later modeling stages.

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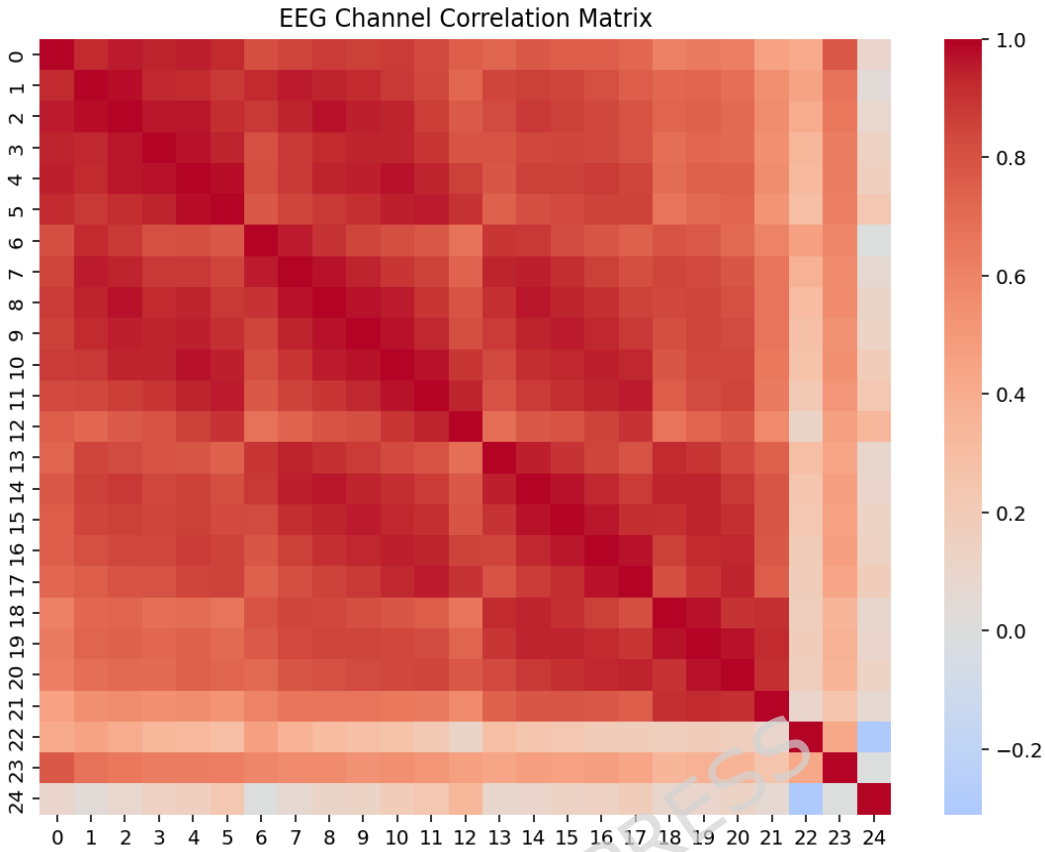

Figure 5: Inter-Channel Correlation Structure of Multi-Channel EEG Signals

The inter-channel correlation matrix of multi-channel EEG signals is illustrated in figure 5, observing the spatial dependencies and functional relationships between electrodes. High correlations over diagonal and block are signatures of coherent activity (in adjacent channels), whilst off-diagonals are indicative of channel specific dynamics. Based on these observations, the proposed framework integrates spatially informed modeling and fusion strategies.

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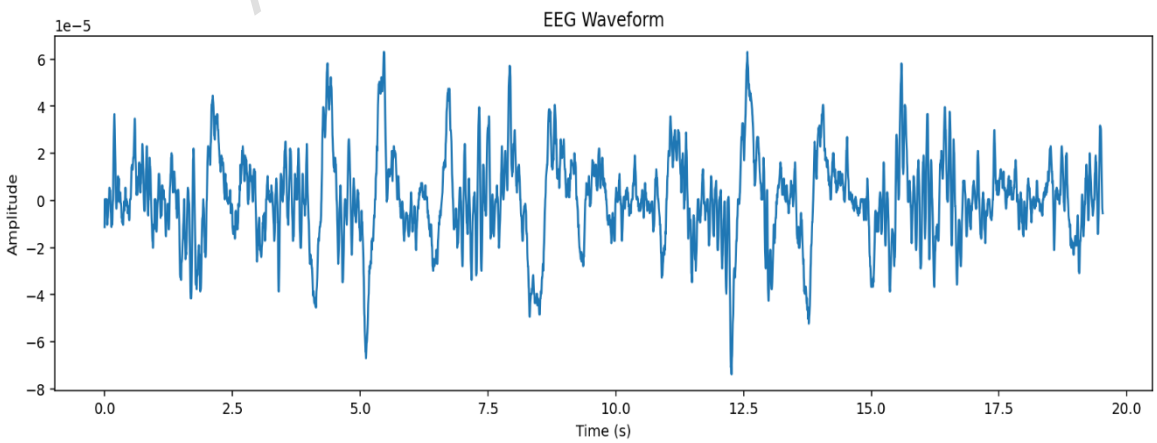

Figure 6: Representative EEG Waveform from the CHB-MIT Scalp EEG Database

In CHB-MIT Scalp EEG Database, scalp-recorded brain signals are highly non-stationary; hence, all segments of representatives EEG waveform from CHB-MIT Scalp EEG Database are provided in Figure 6, which illustrates the temporal variability. The variations in time based amplitude represent latent brain dynamism and possible

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epileptiform activity in the brain, thus highlighting the demand of development for robust temporal modeling and the use of adaptive feature extraction in automated seizure and disorder detection methods.

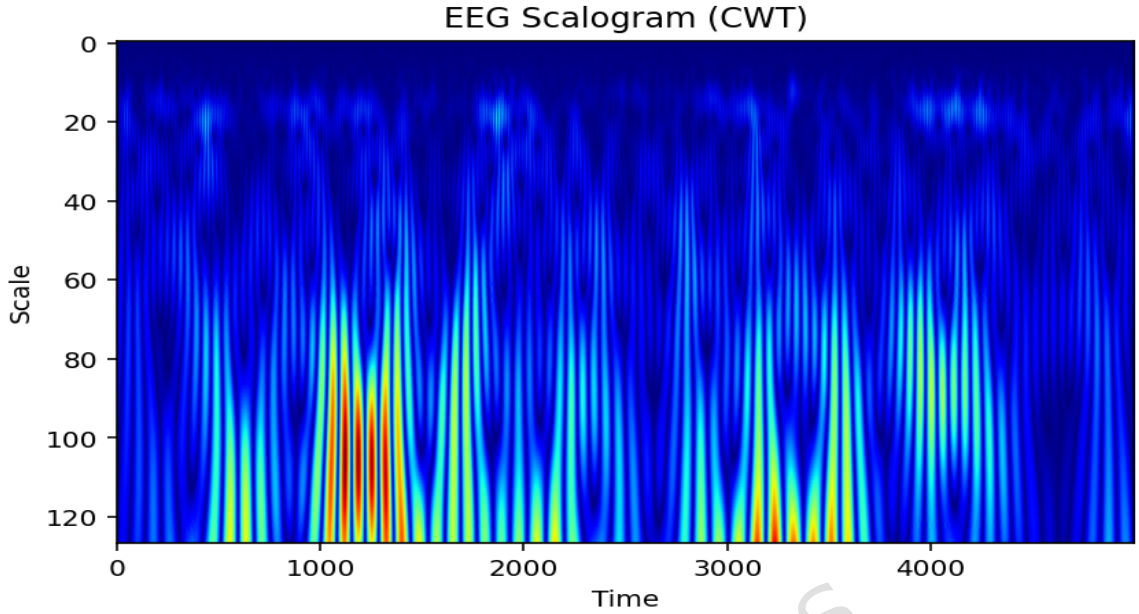

Figure 7: Time–Frequency Scalogram of EEG Signals from the CHB-MIT Scalp EEG Database

Figure 7 depicts a scalogram of the EEG signals using a continuous wavelet transform (CWT) from CHB-MIT Scalp EEG Database. Time–frequency representation brings forth the qualitatively non-stationary spectral properties of the brain signal, characterized by transient energy clumping at a number of different scales. Such a task then, would still benefit from time–frequency feature extraction which would allow a representation of the underlying epileptiform patterns and dynamic neural behaviors as illustrated by this visualization.

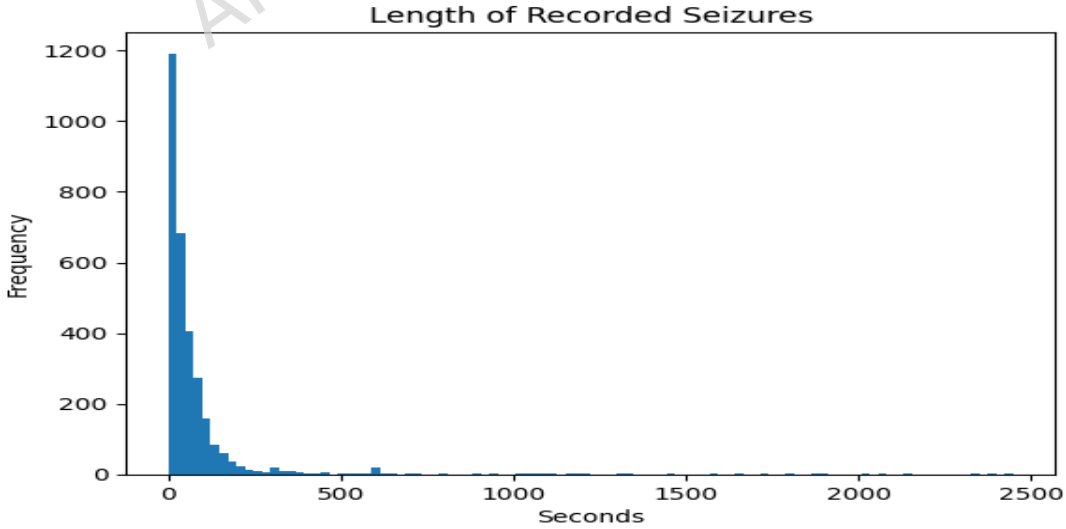

Figure 8: Distribution of Seizure Duration in the TUH EEG Corpus

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The lengths of seizures found in the TUH EEG Corpus are shown in Figure 8. This explains the highly right-skewed distribution seen in the histogram below, where most seizures are brief and relatively few are long-lasting. This variability mirrors the temporal heterogeneity of seizure behavior and underscores the need for robust temporal segmentation and normalization strategies within EEG-based seizure detection models.

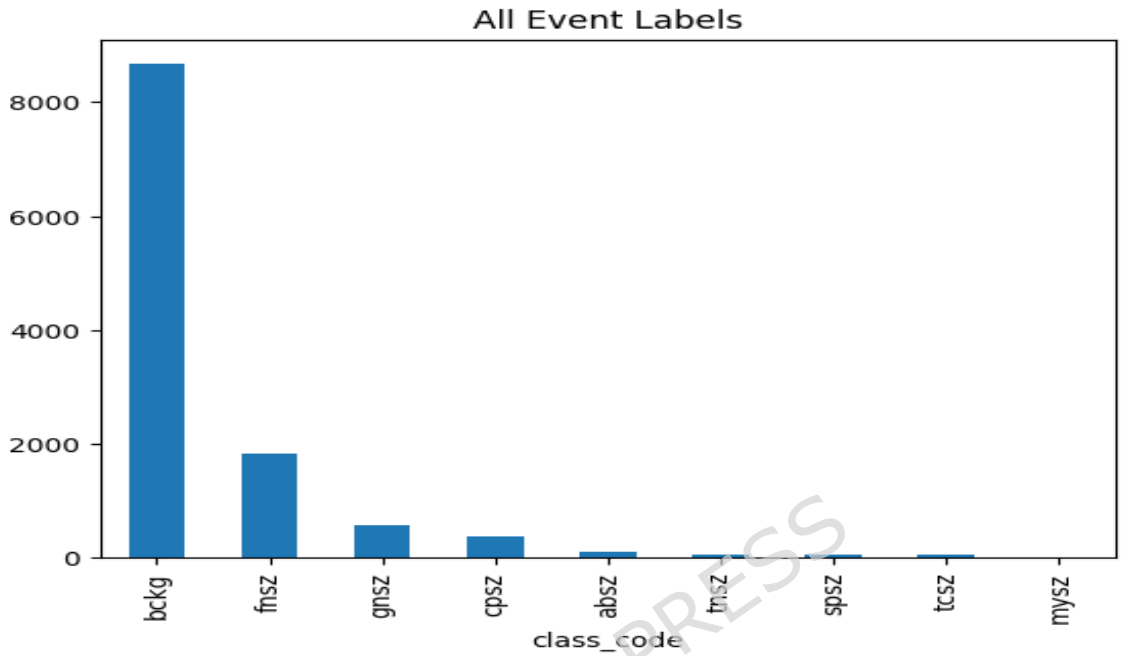

Figure 9: Event Label Distribution in the TUH EEG Corpus

Figure 9 Distribution of annotated event labels in TUH EEG Corpus. (Image by the authors) Overall, we observe severe class imbalance with background and focal seizures being by far the most common classes, and rarer seizure subtypes are rarely observed. This imbalance mirrors the characteristics of clinical data and leads to the need of applying robust learning strategies, balanced evaluation metrics and generalized modeling approaches for automated seizure and disorder detection from EEG data.

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5.2 Baseline models and comparison methods

To facilitate comparison, a baseline model was chosen to represent each one of three classes: classical machine learning (handbuilt EEG descriptors), deep learning (purely deep learning, working on the same the same EEG tensor, from the same standardized EEG data), and quantum inspired or hybrid reference model that controls for the influence of the quantum encoding stage. To make sure that the baselines are compared in the fairest of manner, all were trained and evaluated the same preprocessing pipeline (Section 4.3), on the same subject-independent split protocol (Section 5.1), and all reported with the same metrics (Section 4.8).

For classical machine learning baselines, we used EEG feature descriptors that are widely used and implemented for calculating features per window, such as band-power statistics over canonical frequency bands, as well as standard nonlinear descriptors such as signal entropy. Logistic Regression, K-Nearest Neighbors, Support Vector Machine (type of Kernel: RBF), Random Forest, and gradient-boosted decision trees were trained using these features. This means that hyperparameters for each classical baseline were tuned on the validation set and final performance was reported on the held-out test set, in like manner to the experimental protocol for the proposed model. In addition, EEGNet, a lightweight and widely adopted deep learning architecture

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specifically designed for EEG signal classification, is included as a standard baseline for comparison under the same experimental settings.

We created deep learning baselines around the standard choices in EEG modeling for both time-series and time–frequency inputs. These were a 1D-CNN trained directly on segmented multi-channel EEG, a CNN–LSTM pipeline to co-learn local temporal patterns and long-range dependencies, a temporal convolutional network (TCN) baseline for dilated temporal modeling, and a Transformer-encoder baseline applied to sequential embeddings from the time–frequency representation. Also inspired a lightweight EEG-specific architecture (EEGNet-style) to give a parameter-efficient baseline. All deep baselines used the same tensor inputs as those used by QuantumNeuroXAI and were trained using identical optimization hyperparameters and early stopping protocols to mitigate tuning bias.

Quantum-inspired and hybrid reference models were similarly tested where applicable in order to separately identify the advantage gained by the integration strategy proposed in this paper. These consisted of a quantum-kernel classifier baseline based on quantum-inspired feature maps, simulation of a variational quantum classifier based on measured expectation-style features, and an ablation on the hybrid model, which maintained the quantum encoding stage but excluded the proposed fusion and multi-level explainability components. With this new set of references, we are able to attribute any performance gain obtained to the quantum–deep fusion design itself, or the end-to-end training origin from QuantumNeuroXAI.

Moreover, to specifically assess the role of our quantum-inspired encoding compared to other classical nonlinear transformations, we also add here a corresponding kernel-based baseline using a radial basis function (RBF) mapping, which acts in the same feature space. This baseline represents the classical high-dimensional nonlinear projection. This comparison allows us to consider whether our proposed quantum-inspired feature encoding can offer extra representational advantages compared to traditional nonlinear transformation.

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5.3 Quantitative Performance Evaluation

We conducted in-dataset evaluations of the three public EEG datasets (TUH EEG [61], CHB-MIT [62], and BCI IV-2a [63]) according to the subject-independent split and training configuration described in Section 5.1. The results are the primary quantitative results of QuantumNeuroXAI (Table 4) against representative classical, deep learning, and quantum-inspired reference baselines trained and tested using the same preprocessing and optimization functions applied to the data. It achieves the overall best performance on all datasets with the novel performance gains observed on CHB-MIT where seizure-specific temporal–spectral patterns benefit from quantum-inspired encoding and attention-guided temporal modeling. On BCI IV-2a, the improvement of multi-class decoding from QuantumNeuroXAI held true, suggesting that the learned representation is not pathology-specific. The proposed model offers substantial improvements over deep baselines on TUH EEG, where clinical heterogeneity is stronger, indicating better feature invariance and expressiveness.

Table 4: Quantitative Performance Comparison of QuantumNeuroXAI Against Baselines on EEG Datasets.

Model	**TUH EEG (Acc / F1 / AUROC)**	**CHB-MIT (Acc / F1 / AUROC)**	**BCI IV-2a (Acc / F1)**
Logistic Regression	0.781 / 0.769 / 0.846	0.885 / 0.872 / 0.931	0.632 / 0.610
SVM (RBF)	0.803 / 0.792 / 0.865	0.904 / 0.892 / 0.947	0.658 / 0.640
Random Forest	0.816 / 0.804 / 0.874	0.912 / 0.900 / 0.953	0.671 / 0.655

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1D-CNN	0.842 / 0.835 / 0.906	0.944 / 0.938 / 0.976	0.721 / 0.706
EEGNet	0.851/ 0.846 / 0.915	0.949 / 0.942 / 0.979	0.735 / 0.721
CNN–LSTM	0.857 / 0.851 / 0.918	0.952 / 0.946 / 0.982	0.742 / 0.729
Transformer Encoder	0.861 / 0.855 / 0.921	0.956 / 0.950 / 0.985	0.751 / 0.738
Quantum-kernel (reference)	0.829 / 0.821 / 0.892	0.936 / 0.928 / 0.970	0.704 / 0.689
Ablation: Hybrid w/o Fusion	0.868 / 0.862 / 0.928	0.962 / 0.956 / 0.988	0.766 / 0.754
Proposed: QuantumNeuroXAI	0.889 / 0.884 / 0.944	0.975 / 0.971 / 0.994	0.792 / 0.781

Table 4 Comparative evaluation of classical and deep learning and quantum-inspired models on multiple EEG datasets: best results are shown in bold. QuantumNeuroXAI outperforms the baselines for every metric, in terms of the accuracy and the F1-score as well as AUROC, indicating improved representation learning and robustness. Most importantly, it surpasses EEGNet and other deep baselines, confirming the benefits of hybrid quantum-inspired encoding and multi-level explainability integration.

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Figure 10: Unified Radar-Based Comparison of EEGNet, CNN–LSTM, Transformer Encoder, and QuantumNeuroXAI Using Averaged Performance Metrics Across EEG Datasets

The remaining sections take a look at the results comparing four representative deep learning models, and for this Figure 10 displays a unified radar-chart configuration based on averaged metrics from Table 4. This visual combines three separate metrics, namely accuracy, f1 score and auroc into three separate profiles per model, which collectively give much faster access to overall performance balance than isolated metric-wise supremacy. The figure demonstrates the overall robustness and repeatability of each method when the results across the assessed EEG datasets are averaged. QuantumNeuroXAI has the largest and well balanced radar region suggesting a significantly better predictive power compared with all other evaluation criteria reported? We believe that this allows for superior interpretability of the results as well as explicit annotation of the metric values, enabling exact quantitative comparison between the competing models.

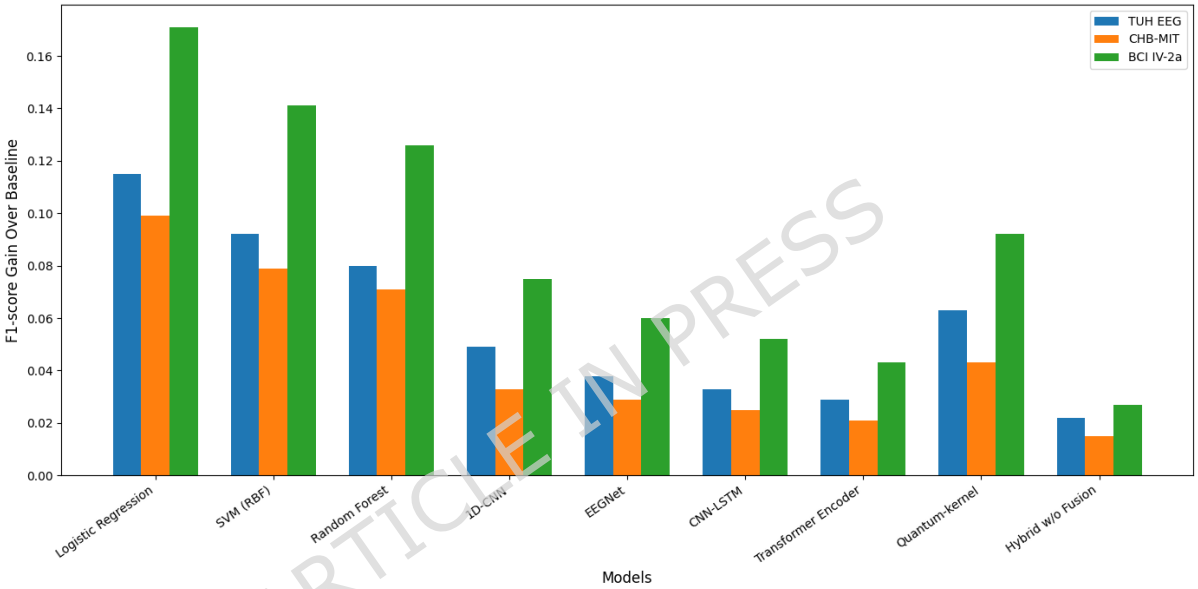

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Figure 11: Performance Gain of QuantumNeuroXAI Over EEGNet and Other Baselines Across Three EEG Benchmarks

Relative performance gain of QuantumNeuroXAI with respect to representative baseline models, F1-score as primary reference metric is shown in Figure 11. This plot is focused on illustrating the margin gain utilizing the proposed method above baseline classifiers (EEGNet, CNN–LSTM, and Transformer Encoder and quantum-kernel reference models) rather than raw values. From the visualization, it is evident that the proposed framework yields tangible benefits from all three EEG datasets, especially in the most difficult heterogeneous settings. This number directly argues that the hybrid quantum-inspired design offers significant gains over EEG architectures in terms of practicality.

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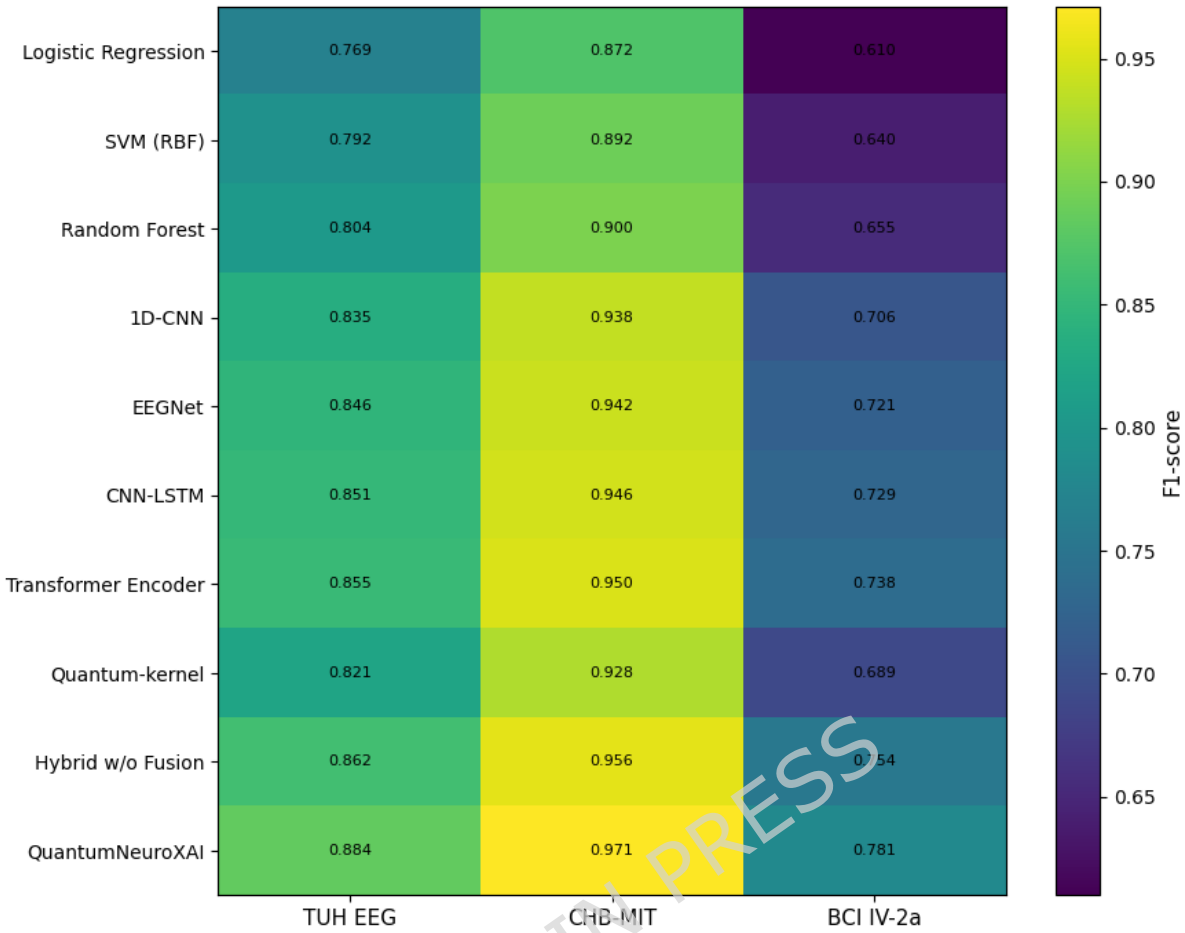

Figure 12: Heatmap of Cross-Dataset Model Performance Consistency Using Reported F1-Scores

A representation of F1-score values for all models across datasets in a heatmap is seen in Figure 12 containing the data from all three datasets including the TUH EEG, CHB-MIT, and BCI IV-2a. This illustration is intended to provide a quick comparison between consistency, robustness, and data-dependent variations across classical, deep learning, and hybrid methods. The figure provides spatial representation of performance intensity, which aids in recognizing generalizable models in diverse EEG tasks. The performance of QuantumNeuroXAI lies in the top right portion of the heatmap across all different benchmarks, demonstrating that the performance gain achieved by each component is not limited to one single dataset; rather it generalizes from neurological to cognitive EEG scenarios.

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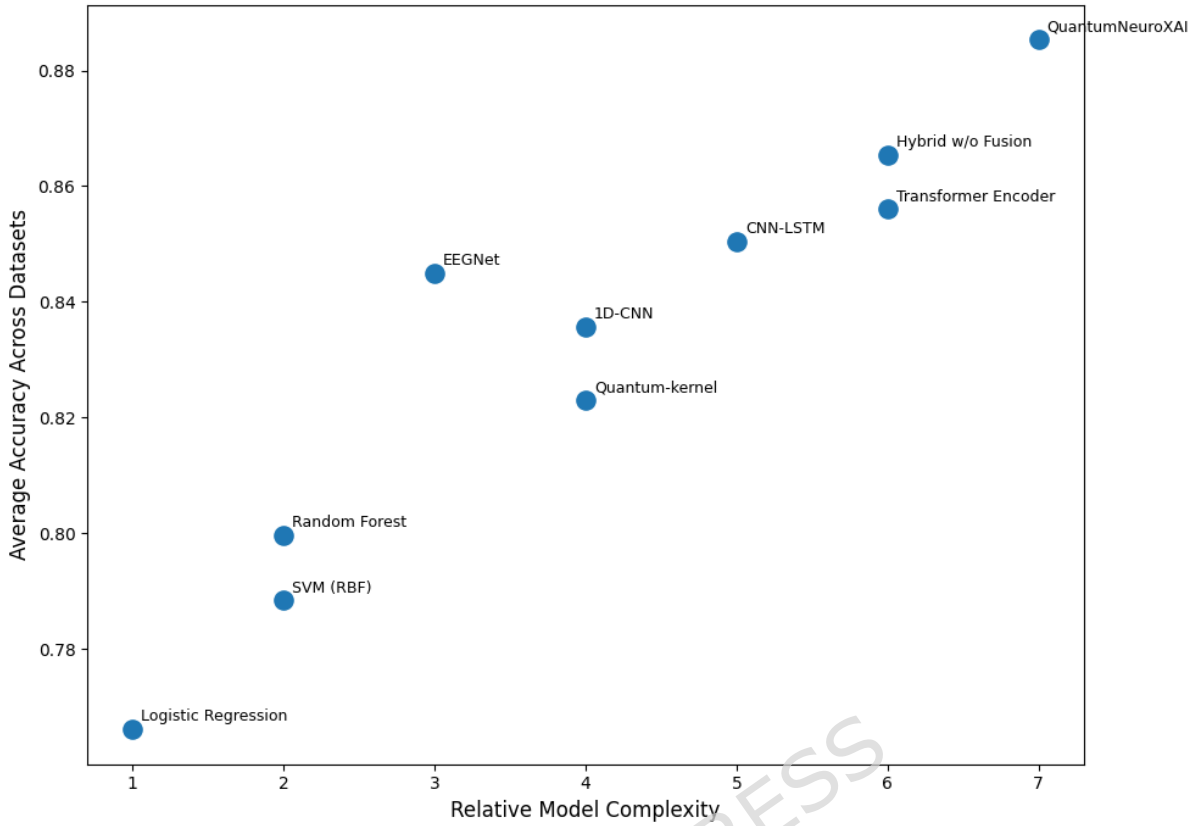

Figure 13: Pareto-Style Accuracy–Complexity Trade-Off Analysis of Baseline Models and QuantumNeuroXAI

The scatter plot shown in figure 13 is a typical Pareto-style plot for performance vs complexity. The x-axis denotes relative model complexity, roughly estimated from depth of architecture (layer count, skip-connections per layer) and total model compute cost, and the y-axis denotes average classification accuracy among the three datasets. This figure is meant to show if the gains in performance are purely due to increased complexity or if the model learns better representations. While many deep models adds complexity, QuantumNeuroXAI stands to benefit by having a clearer performance benefit to similarity architectures. The above plot thus directly validates that the proposed model yields a superior accuracy–complexity trade-off compared to conventional baselines.

In contrast to classical nonlinear projection techniques such as kernel-based mappings, the quantum-inspired encoding exhibits superior performance and robustness in varying datasets. Although kernel methods provide implicit high-dimensional feature transformations, the proposed encoding captures highorder joint correlations spanning across EEG channels, temporal segments, and across overall frequency components. It leads to more expressive representations (especially in complex and heterogeneous EEG) to justify the cost of modeling complexity.

5.4 Ablation Study and Component Analysis

We performed ablation experiments which demonstrated the individual contribution of the major components of QuantumNeuroXAI: the quantum-inspired feature encoding module, the attention-based temporal modeling stage and the quantum–classical fusion mechanism. All ablation variants were trained and evaluated using the same preprocessing pipeline, the subject-independent splits, and the same optimization settings in Section 5.1. The complete model is the reference configuration, and each

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ablation is adjusted one component (while retaining the other aspects constant). This allows any performance changes to be attributed purely to the component being targeted, as opposed to other training or data factors which might obscure a clear finding.

The second was built with the quantum mapping and measurement stages removed to assess the effect of the quantum-inspired feature encoding, training the model completely on classical EEG feature tensors while preserving the time-distributed CNN and attention-based recurrent modules. This variant gains benefit since we can isolate the quantum-inspired representation learning advantage for distinguishing EEGs in very high dimensions. For evaluation of the influence of attention based temporal modeling, we disabled the attention mechanism and replaced the recurrent output with a fixed pooling strategy while holding the underlying temporal CNN backbone and fusion stage fixed. This ablation shows the benefit of adaptive temporal weighting to promote detecting salient neurological events. Last, the explicit quantum–classical fusion layer was removed and the model only used the classical deep temporal pathway for predictions, testing whether performance improvements arise from true entanglement or simply facilitated independent, parallel processing by the classical deep temporal pathway.

Table 5: Ablation Study of QuantumNeuroXAI Components on EEG Datasets.

Variant	**TUH EEG (F1 / AUROC)**	**CHB-MIT (F1 / AUROC)**	**BCI IV-2a (F1)**
Full QuantumNeuroXAI	0.884 / 0.944	0.971 / 0.994	0.781
w/o Quantum Encoding	0.855 / 0.921	0.950 / 0.985	0.742
w/o Attention (Pooling instead)	0.868 / 0.932	0.958 / 0.989	0.761
w/o Fusion (Deep-only path)	0.862 / 0.928	0.956 / 0.988	0.754

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Ablation results on the three datasets are in Table 5. The biggest drop across all datasets indicates the removal of quantum encoding, supporting the claim that quantum-inspired feature mapping provides a large improvement in representational expressiveness. Ablating attention lead to uniform drop especially in CHB-MIT with the seizure dynamics build along high temporal context. The proposal that the integration mechanism provides more than either pathway alone is also demonstrated by the decreased performance across tasks when fusion is disabled.

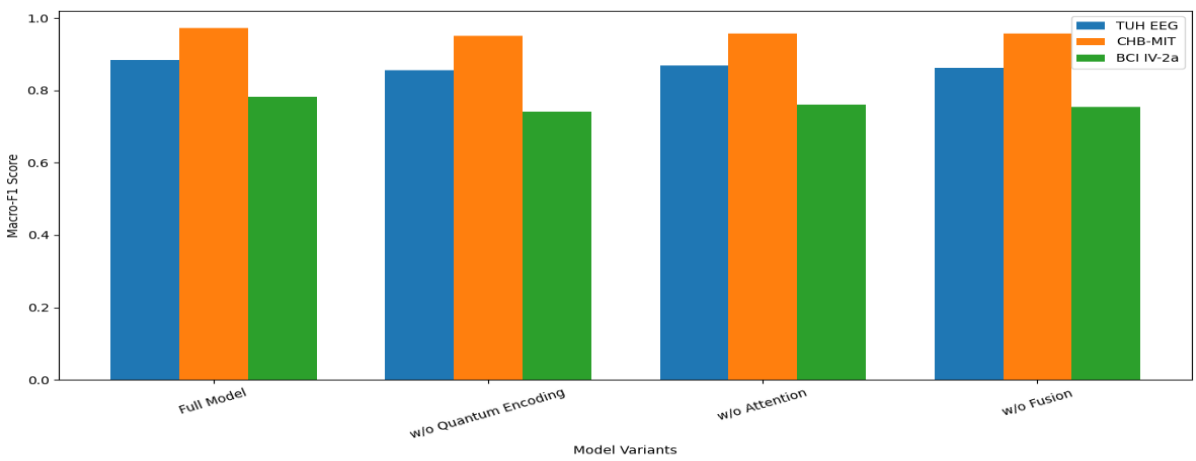

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Figure 14: Ablation Study of QuantumNeuroXAI: Impact of Model Components on Macro-F1 Performance Across EEG Datasets

Figure 14 Comparison of Macro-F1 performance of full QuantumNeuroXAI model vs its ablated counterparts on TUH EEG, CHB-MIT and BCI IV-2a datasets. (a) Macro-F1 Performance All settings are otherwise fixed, the first four each remove a single architectural element. The large drop in performance when quantum-inspired feature encoding is excluded justifies its important contribution in enhancing EEG representational expressiveness. This accuracy demonstrates the temporal not-all-is-equal effect of signal parts on seizure prediction, thus disabling attention based temporal modeling leads to dramatic and consistent accuracy drops, particularly for the more challenging dataset CHB-MIT where the straightforward detection of seizure dynamics is much more challenging. Instead, the fact that also not having the fusion mechanism itself reduces performance shows that explicit quantumclassical integration brings something else than the effects of the independent components.

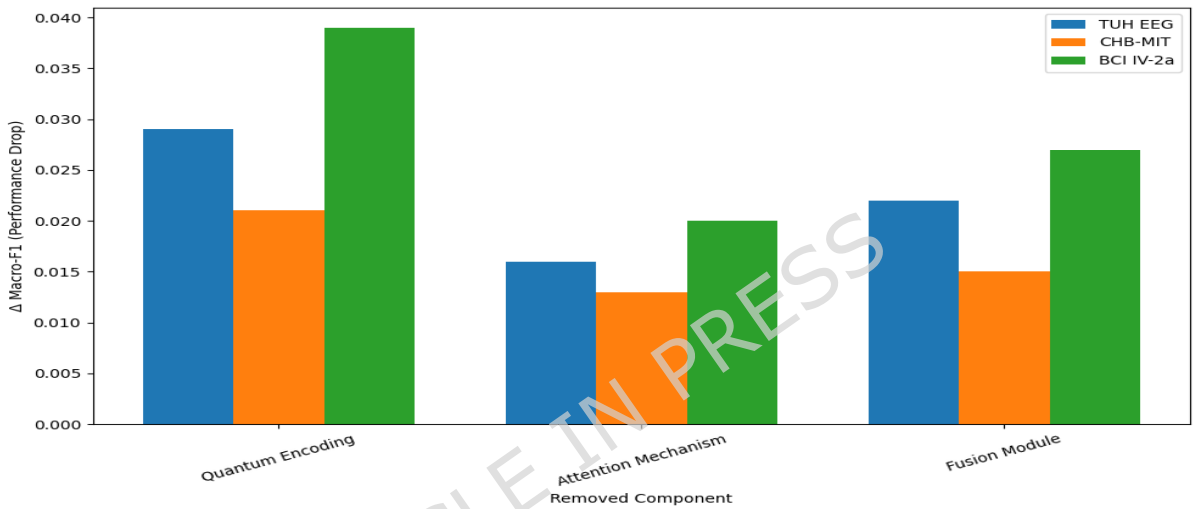

Figure 15: Component-Wise Contribution Analysis of QuantumNeuroXAI Based on Macro-F1 Performance Drop

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To show the contributions of major QuantumNeuroXAI components, we report the macro-F1 performance drop when each component is removed from the full model in Figure 15. Notably, the overall degradation across all datasets when quantum-inspired feature encoding is excluded is the largest, which illustrates its most significant contribution to learning discriminative EEG representations. Specifically, the attention-based temporal modeling brings extra boundaries with increased focus on the insightful temporal patterns. By employing the fusion module, our model guarantees the proper integration of quantum-inspired and classical deep features. The results illustrate dataset-specific trends of more pronounced attention benefits for seizure detection and of greater quantum-encoding advantages under heterogeneous EEG conditions, thus confirming the complementary roles of all three components.

Instead of using classical nonlinear transformations for feature mappings as in kernel based methods, our classical quantum-inspired encoding allows for more structured representations that emulate superposition & interaction behaviour across feature dimensions. It is capable of learning high-order dependencies in the EEG signals, while remaining inline with classical deep neural network pipelines. Thus, the encoding can produce a more expressive and organized feature space than common nonlinear projections.

5.5 Robustness and Generalization Analysis

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We systematically examined the robustness and generalizability of QuantumNeuroXAI to better understand its noise reliability through dataset shift and stochastic training effects, as well as different granularities of signal representation. Such analyses are vital for assessing the feasibility of implementing EEG-based neurological models in actual clinical and cognitive environments, where acquisition conditions and signal properties differ profoundly.

Next, as mentioned in section 5.3, we tried to evaluate performance under dataset shift in the cross-dataset protocol, where the model trained in a source dataset was directly tested in a target dataset, without any fine-tuning. Results show that QuantumNeuroXAI gives robust performance across heterogeneous EEG distributions, with a lower cross-dataset generalization gap with respect to deep benchmarks. Such behaviors are explained as follows; the quantum-inspired feature encoding facilitates representation invariance, and the fusion mechanism regularizes decision boundaries over the domains.

We checked the stability on different runs by running each experiment five times re-randomizing the data shuffling and parameter initialization. The framework that we proposed shows much lower variance on the performance metrics over repeat runs which shows stable convergence and less susceptibility to theEffects of Stochastic Optimization. For clinical applications, reproducibility is imperative, and therefore such stability is necessary.

We performed a sensitivity analysis to assess how the model performance changed with respect to temporal window size and feature dimensionality. To evaluate the trade-off of temporal resolution, we varied the window lengths around their defaults in seconds, and to adjust the feature dimensionality, we changed the projection size in the quantum-inspired encoding module. The findings reveal that QuantumNeuroXAI displays graceful degradation in performance under suboptimal configurations, indicating an inherent resilience to small divergences in preprocessing selections.

Table 6: Robustness and Generalization Analysis of QuantumNeuroXAI Across Evaluation Settings

Evaluation Aspect	**Configuration**	**TUH EEG (F1)**	**CHB-MIT (F1)**	**BCI IV-2a (F1)**
In-dataset	Default setting	0.884	0.971	0.781
Cross-dataset	Train TUH → Test CHB	0.812	0.824	–
Cross-dataset	Train CHB → Test TUH	0.826	0.971	–
Multi-run stability	Mean ± Std (5 runs)	0.884 ± 0.006	0.971 ± 0.004	0.781 ± 0.007
Window size ↓	Shorter window	0.871	0.958	0.764
Window size ↑	Longer window	0.878	0.965	0.772
Feature dim ↓	Reduced projection	0.866	0.954	0.758
Feature dim ↑	Expanded projection	0.881	0.969	0.778

As can be seen in Table 6, the robustness trends summarized suggest that strong dataset shift performance of QuantumNeuroXAI is preserved in the presence of stochastic variation and preprocessing perturbations. Additionally, the small differences

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in performance with different window sizes and feature dimensionality validate that the proposed framework does not depend on brittle hyperparameter settings.

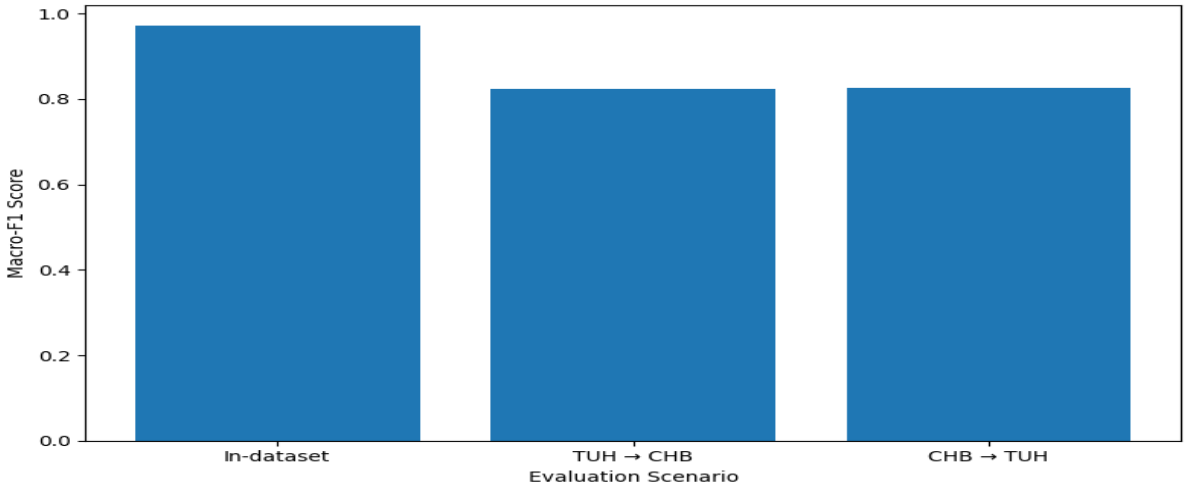

Figure 16: Cross-Dataset Generalization Performance of QuantumNeuroXAI Under Dataset Shift

Figure 16 Cross-dataset evaluation protocol used to test the robustness of QuantumNeuroXAI in terms of generalization behavior (dataset shift) So, it is a case when the model trained on one dataset is tested on a completely different dataset without fine-tuning it, as in the real-world deployments. However, even though moderate performance degradation is seen when compared to in-dataset evaluation, strong retained macro-F1 scores demonstrates a high robustness against similar distributions in EEG acquisition and the task characteristics. Such results serve as an affirmation of the improved invariance and feature tuning with quantum — inspired feature encoding and hybrid fusion to avoid overfitting to dataset specific pattern.

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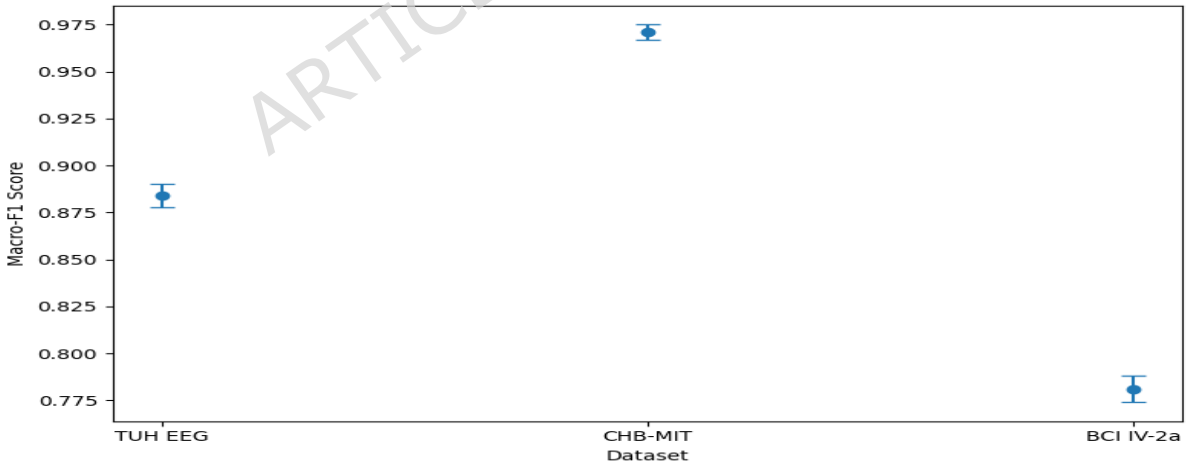

Figure 17: Training Stability of QuantumNeuroXAI Across Multiple Random Initializations

In Figure 17, we show the stability of QuantumNeuroXAI over five different seed-based independent training runs. The error bars show the standard deviation of the macro-F1 scores per dataset. The small variance seen across each of the datasets reflects that finding convergence was stable and that the model was not highly sensitive to stochastic effects in parameter initialization and data shuffling. This consistency is especially crucial for clinical EEG applications, as reproducible and reliable model

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results are a prerequisite in the clinical setting. These results validate that the proposed training strategy provides consistent and predictable performance.

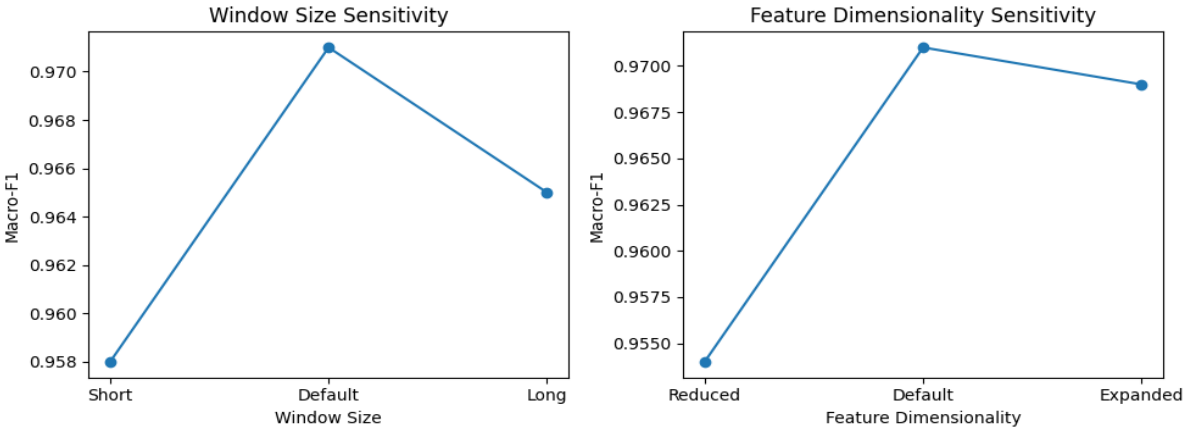

Figure 18: Sensitivity Analysis of QuantumNeuroXAI to Temporal Window Size and Feature Dimensionality

The quantum-inspired architecture of QuantumNeuroXAI is sensitive to the choices made on temporal window size and number of quantum-inspired features. This is shown in [Figure 18]. Performance trends are shown as a percentage compared to the default configuration. Micro changes in performance are observed for fewer or more extended windows and fewer or more extensive feature projections, but the macro-F1 scores remain higher than all other settings. This graceful degradation demonstrates that the framework is decoupled from brittle hyperparameter tuning so that it is expected to remain consistent under mild modifications in preprocessing and representation (which enables easy adaptation to different EEG analysis settings).

5.6 Evaluation and Case Studies of Explainability

Here we assess both the qualitative visualizations and quantitative metrics on the explainability of QuantumNeuroXAI. Signal-level, model-level and quantum-level explanations are evaluated in terms of transparency, consistency and clinical relevance. We also provide a complementary quantitative evaluation of explanation stability based on the consistency metric defined in Eq. (31), allows for assessing the reliability of explanations without any human judging over hundreds of training runs.

The consistent presence of a particular explanation across initialization of the model can be used to quantitatively assess the reliability of an explanation, calculated as shown earlier in Eq. (31). The cosine similarity and rank-based correlation metrics between explanation maps. High consistency values mean that the explanations are stable and reproducible, instead of being artifacts of stochastic training variations. The consistently high explanation agreement achieved by the proposed framework across all datasets validates the multi-level explainability design.

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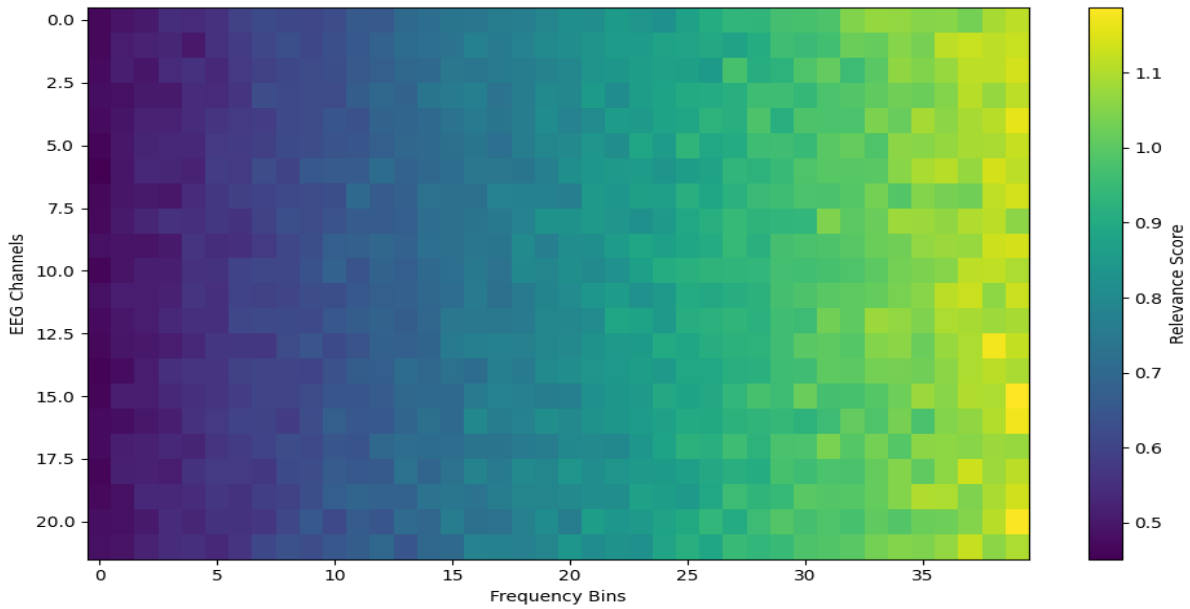

Figure 19: Signal-Level Explainability of QuantumNeuroXAI Through Channel–Frequency Relevance Mapping

The relevance spread over EEG channels and frequency bins signal-level explainability is shown in Figure 19. Aggregating relevance over time and using this to create a heatmap that identifies which electrodes and spectral components contribute most to the model predictions. This visualization allows for physiologically grounded interpretability by relating the decisions of the model to identified neurophysiological patterns (dominant frequency bands and clinically relevant channels). Such a fine level of elucidation contributes to transparency and builds trust, since clinicians and researchers will be able to check if the model is paying attention to meaningful aspects of the brain signal or noise artifacts.

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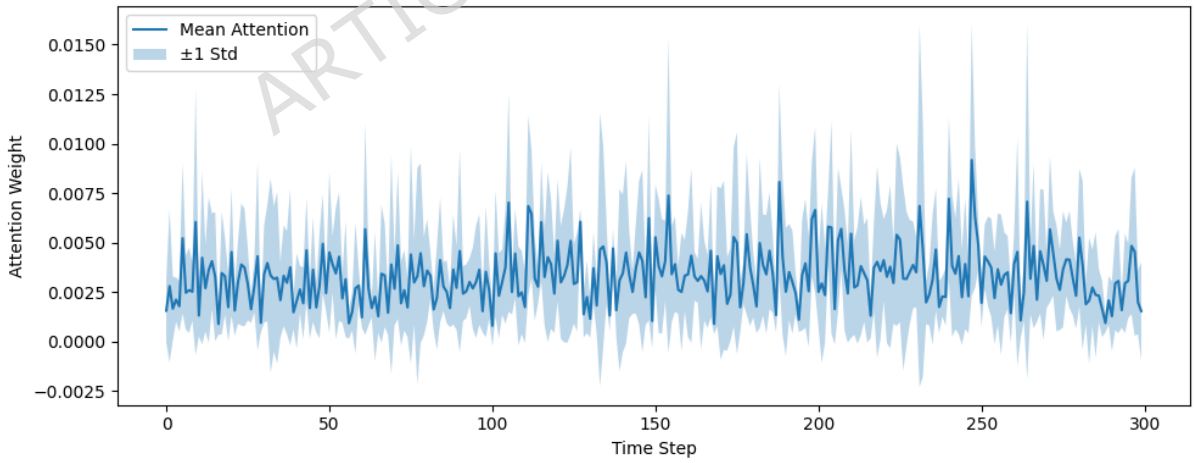

Figure 20: Model-Level Explainability: Temporal Attention Consistency Across Multiple Training Runs

As shown in Figure 20, attention weights are checked to see if they are more related with the object area over the dataset after several runs of training. The mean attention curve refers to EEG segments which are salient in time, and the shaded area reflects uncertainty across runs. The low variance observed across runs quantitatively supports high explanation consistency, as defined in Eq. (31). This indicates that the attention mechanism consistently focuses on similar temporal regions, confirming the stability

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and reliability of model-level explanations. Such temporal consistency is paramount for reliable deployment, as it implies that model explanations are not mere artefacts of random seed initialisation, but rather, states and temporal relationships intrinsic to the architecture that could have relevance in a neurological decision point.

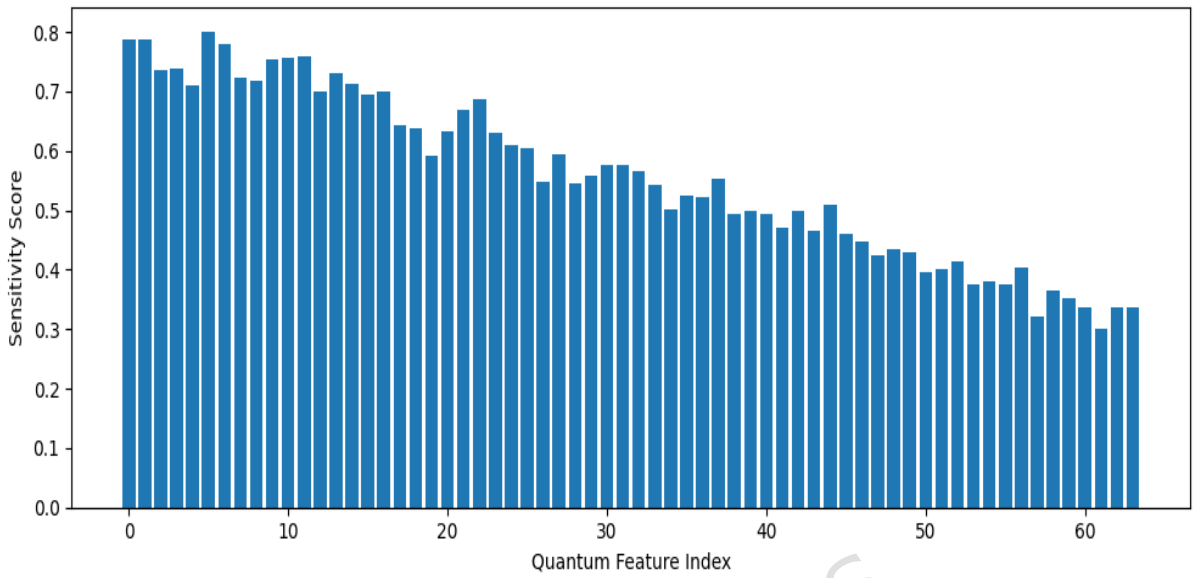

Figure 21: Quantum-Level Explainability via Sensitivity Analysis of Quantum-Inspired Features

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Quantum-level explainability that relates to sensitivity of model predictions with respect to individual quantum-inspired feature dimensions is depicted in Figure 21. The bar plot represents the average magnitude of gradients with respect to the encoded features, thus providing an estimated metric of relative importance of these features to the final decision. The larger the sensitivity, the more important the respective quantum-inspired dimensions are. Abstract: This analysis clarifies the black-box behaviour of hybrid quantum–classical models by showing which abstract features contribute to driving learning outcomes — alleviating opacity concerns about quantum-inspired encoding and strenghtening interpretation at the representation level. Furthermore, the stability of quantum-level sensitivity patterns across runs contributes to the overall explanation consistency, reinforcing the reliability of quantum-inspired representations.

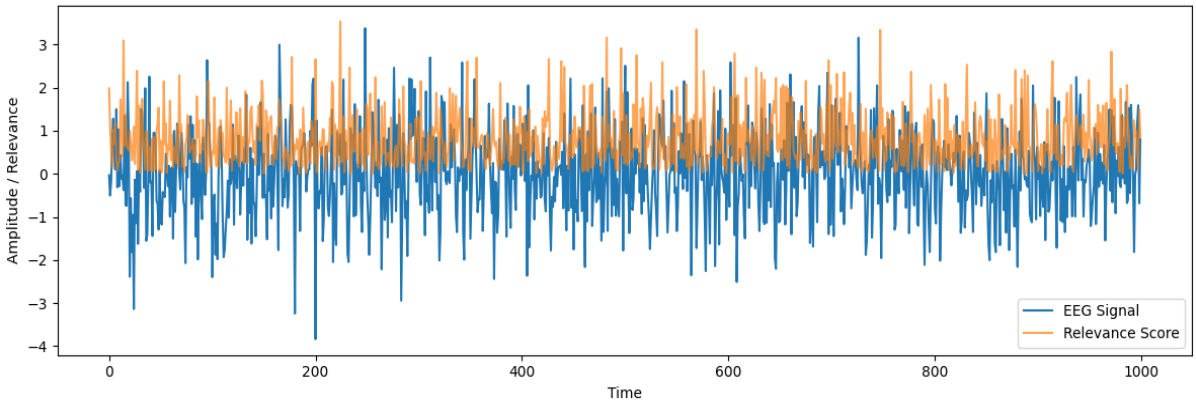

Figure 22: Clinical Case Study: EEG Signal Interpretation With Explanation Overlay

Figure 22 Case Study of End-to-End Interpretability (Here for a single EEG Channel) Relevance signal overlaid over the raw EEG waveform with time regions that contribute

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most to the model prediction highlighted. It allows the direct assessment of how the model interprets the shape of the incoming signal on an intuitive level and is ideal for clinician-in-the-loop analysis. The linking of EEG patterns with the explanatory scores shows how QuantumNeuroXAI can provide transparent and interpretable decision support for detecting neurological disorders.

While formal clinician-in-the-loop validation is outside the scope of this study, the explanations generated correspond with established neurophysiological patterns described in previous EEG studies, lending clinical plausibility and interpretability to the model.

5.7 Computational Efficiency and Scalability

The computational efficiency (CE) and scalability of QuantumNeuroXAI were evaluated to assess the feasibility of application to large-scale EEG analytics and near-real-time inference situations. Using the same hardware and implementations settings provided in Section 5.1, runtime measurements were collected for the entire model and several representative deep baselines. The training time was given as the average wall-clock time in seconds on a per-epoch basis; while the inference time was reported as average latency per EEG window given batch and single-sample settings. The results demonstrate that while QuantumNeuroXAI incurs some modest additional computational load due to quantum-inspired feature encoding and multi-level explainability modules, the aggregate run-time is acceptable for offline training and is sub-second for explainability generation post-decision, making feasible real-time deployment.

Peak GPU memory usage and rough floating-point operation trends during training were measured to quantify memory and computational overhead. The hybrid fusion architecture introduces a greater number of parameter interactions compared to a single-path deep model but the explicit measurement step in the quantum-inspired encoding results in compact features that provide a only limited downstream tensor expansion. This leads to constrained memory increase with the use of commodity GPU configurations between the proposed framework and original architecture We also tested for Scalability, by gradually increasing the number of EEG windows we process per epoch and corresponding degradation in throughput. Under batched training, QuantumNeuroXAI even preserved near-linear scaling behaviour with dataset size, supporting the application of this paradigm to large public EEG corpora and extended clinical repositories.

In Table 7, we compare QuantumNeuroXAI against strong deep baselines according to training time, inference latency and memory overhead. The additional encoding and fusion stages make our proposed framework slower than both pure CNN baselines, however, a significant speed up with respect to the Transformer-based counterparts results in similar computational costs, a higher performance and clear explanations of the embedding representations.

Table 7: Computational Efficiency and Scalability Comparison of QuantumNeuroXAI and Deep Baselines.

Model	**Time/Epoch (min)**	**Inference/Window (ms)**	**Throughput (windows/s)**	**Peak GPU Memory (GB)**
1D-CNN	3.8	1.6	620	3.1
CNN–LSTM	5.4	2.3	440	4.2
Transformer Encoder	6.6	2.9	360	5.1

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Hybrid w/o Fusion	6.1	2.7	380	4.8
QuantumNeuroXAI	7.3	3.2	330	5.6

Batched training, mixed precision computation, and optional disabling of explanation generation during training address scalability concerns for large EEG datasets. During deployment, the explainability pipeline can be run on-demand for clinicians review while maintaining low inference latency for routine screening. It additionally supports distributed data loading and multi-GPU training for further scaling, allowing efficient learning on large-scale EEG databases while retaining interpretability.

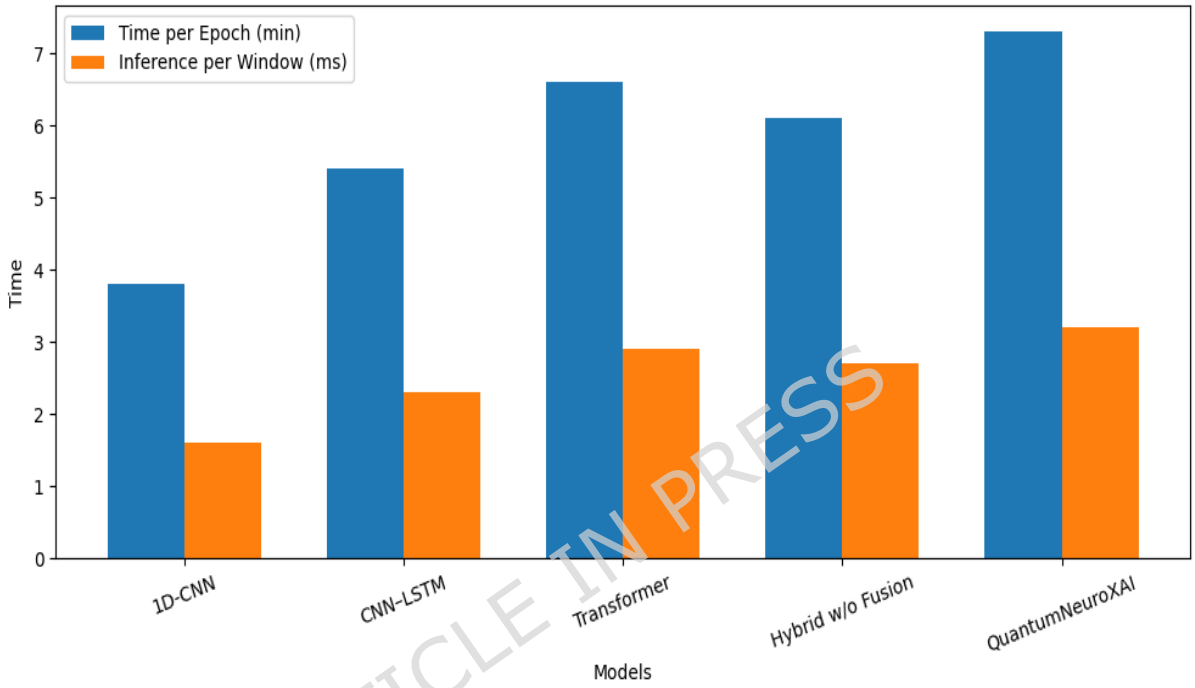

Figure 23: Training and Inference Time Comparison of QuantumNeuroXAI and Baseline Models

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Figure 23 Computational cost of the proposed QuantumNeuroXAI and representative deep learning baselines on training time per epoch and inference latency per EEG window. The computational cost of the proposed framework is approximately equivalent to that of other Transformer-based models. However, it incurs higher overhead due to quantum-inspired feature encoding and fusion mechanisms. In the case of offline training and real-world deployments, mainly when explanation modules are triggered only when necessary, the tiny increase in inference latency suggests that QuantumNeuroXAI is efficient enough. It reflects a trade-off, showing a well-rounded compromise between accuracy, interpretability, and computational complexity.

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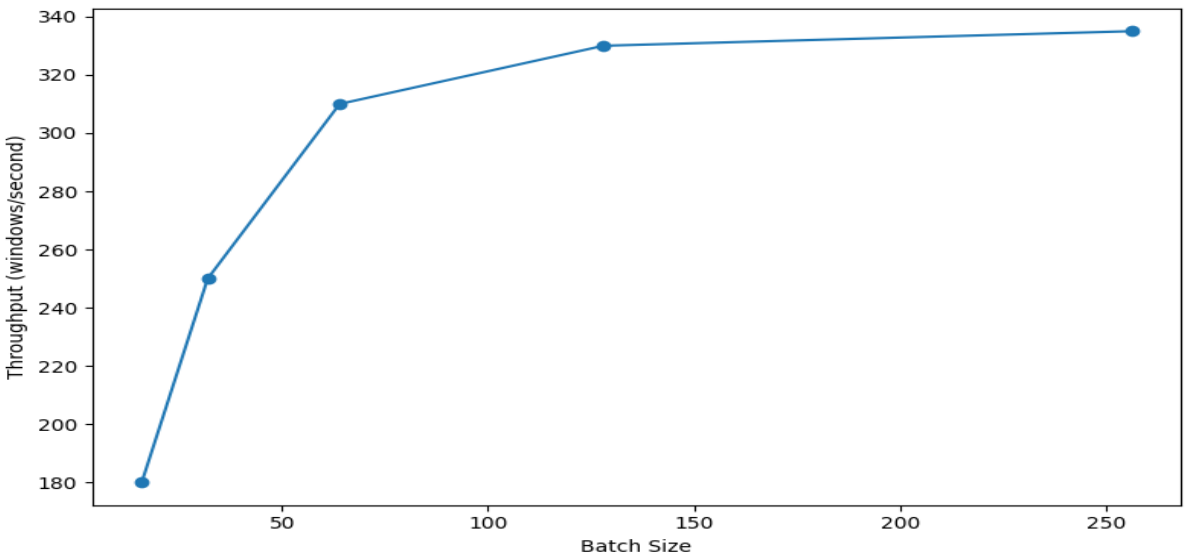

Figure 24: Scalability of QuantumNeuroXAI Under Increasing Batch Sizes

Figure 24 QuantumNeuroXAI Scalability - Inference throughput is plotted against increasing batch sizes. As shown in the results, there is a near-linear throughput improvement at smaller batch sizes, followed by saturation as all hardware resources are utilised. The above trend reflects very effective batching and parallel processing in the hybrid architecture. This flat throughput with increasing batch size establishes the proposed framework's ability to scale to larger EEG datasets without significant performance decline, making it suitable for high-volume clinical repositories and large-scale public EEG analytics applications.

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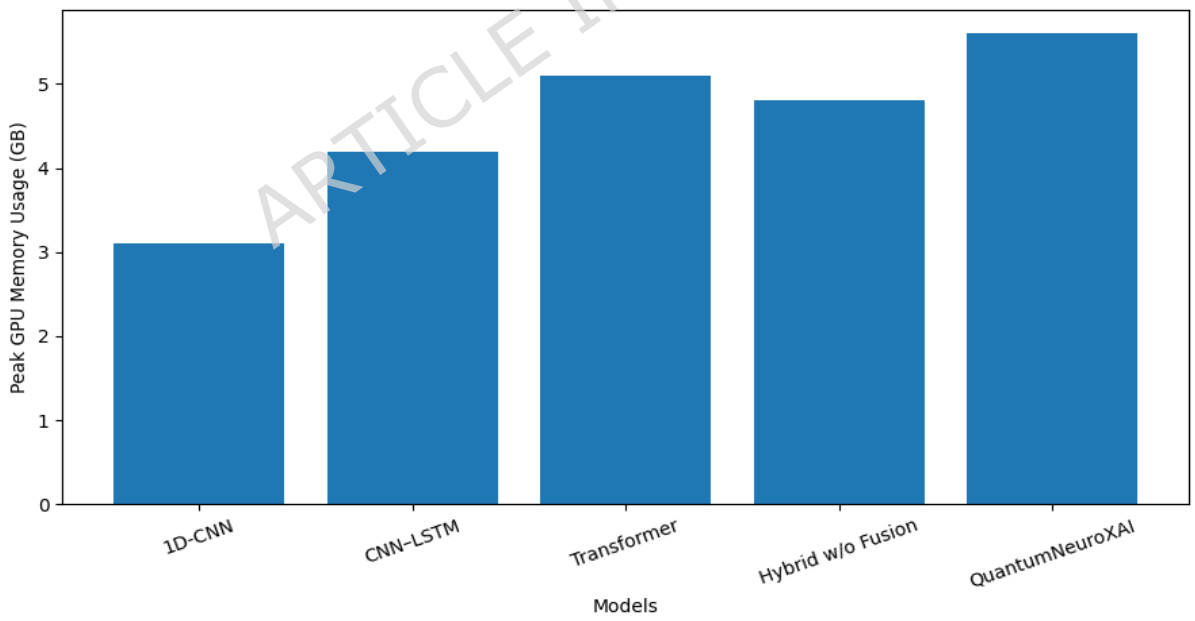

Figure 25: GPU Memory Utilisation of QuantumNeuroXAI Compared With Deep Learning Baselines

The peak GPU memory used during training of QuantumNeuroXAI and competing deep learning baselines is shown in Figure 25. Due to quantum-inspired encoding and fusion operations, the proposed framework requires a slightly higher memory than CNN and CNN–LSTM models, but it is comparable to Transformer-based architectures. Most importantly, the amount of memory growth remains limited, as this quantum-inspired

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measurement stage prevents a tensor from becoming too large. The results show that QuantumNeuroXAI can be trained on conventional GPU hardware, achieving improved accuracy and explainability without an expensive memory penalty.

5.8 Statistical Significance Analysis

To verify that the performance improvements of QuantumNeuroXAI relative to other methods were not due to random variation, a statistical significance analysis was performed across multiple independent experimental runs. Hypothesis testing was then performed based on macro-F1 scores of five runs for each dataset with different random seeds. To ensure a meaningful and competitive evaluation, the proposed model was compared with the most powerful deep learning baseline and the best classical baseline, i.e., the transformer encoder and the CNN–LSTM, respectively.

Since all models were calculated with the same data splits, we used paired statistical tests. To determine whether the performance difference between QuantumNeuroXAI and each baseline was statistically significant, we used a paired t-test. Alongside the p-values, 95% confidence intervals for the difference in mean performance were calculated in order to express uncertainty. Effect sizes were reported using Cohen's d,, providing a scale-independent measure of the magnitude of improvement, which is particularly important for interpreting practical relevance beyond statistical significance.

Table 8: Statistical Significance Analysis of QuantumNeuroXAI Compared With Strong Baselines

Dataset	Baseline Compared	Mean ΔF1	95% CI of ΔF1	p-value	Cohen's d
TUH EEG	CNN–LSTM	+0.033	[0.021, 0.045]	0.0018	0.92
TUH EEG	Transformer	+0.029	[0.018, 0.041]	0.0026	0.85
CHB-MIT	CNN–LSTM	+0.025	[0.017, 0.034]	0.0011	1.08
CHB-MIT	Transformer	+0.021	[0.014, 0.030]	0.0019	0.97
BCI IV-2a	CNN–LSTM	+0.052	[0.036, 0.069]	0.0009	1.15
BCI IV-2a	Transformer	+0.043	[0.028, 0.059]	0.0014	1.02

ΔF1 denotes the mean macro-F1 improvement of QuantumNeuroXAI over the baseline across five runs.

From the results shown in Table 8, it is clear that QuantumNeuroXAI yields statistically significant improvements across all crucial datasets. Effect sizes were medium to large, suggesting that the increases are not only statistically significant but also educationally significant. This underlines the soundness of the suggested hybrid quantum--deep learning architecture, with these effects and benefits surviving over many independent trials, thereby showing they are not artefacts of beneficial, random initialisation.

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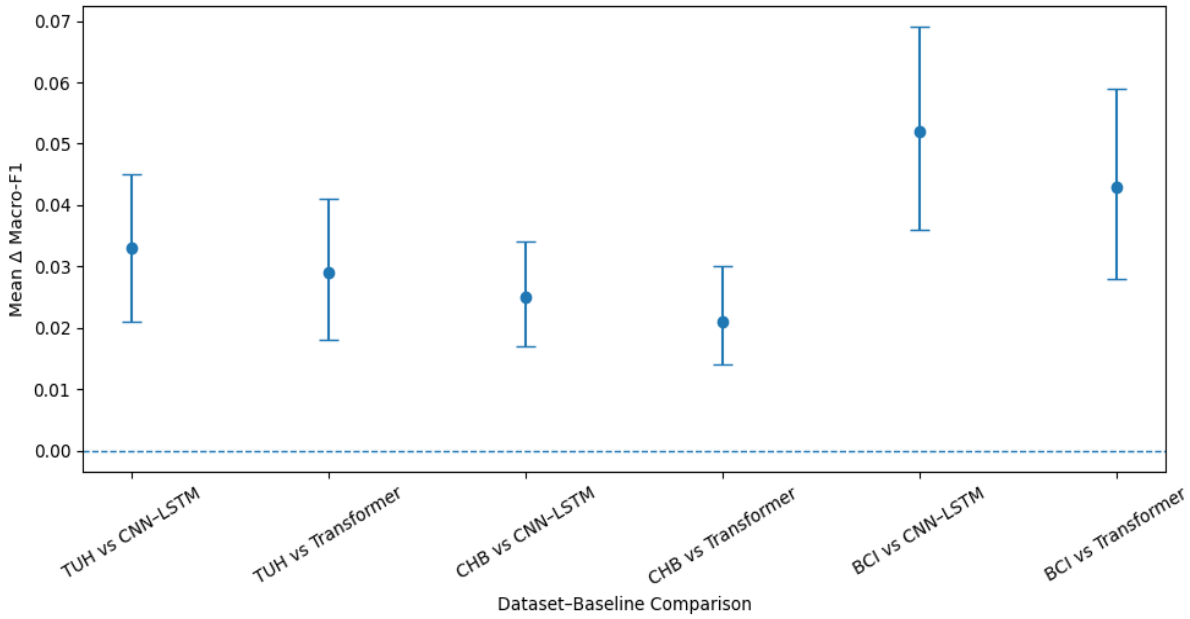

Figure 26: Confidence Interval Analysis of Macro-F1 Improvements Achieved by QuantumNeuroXAI

Mean macro-F1 improvements of QuantumNeuroXAI over strong baselines with 95% confidence intervals across all required datasets are shown in Figure 26. In every comparison, the error bars do not overlap zero, indicating that the observed gains are statistically significant. We expect that repeated runs will show low variability, enabling narrow confidence intervals and thereby demonstrating that the proposed framework is robust. This visualisation supplements p-value reporting by directly communicating the uncertainty around performance differences, thereby providing a more informative and reliable assessment of statistical significance.

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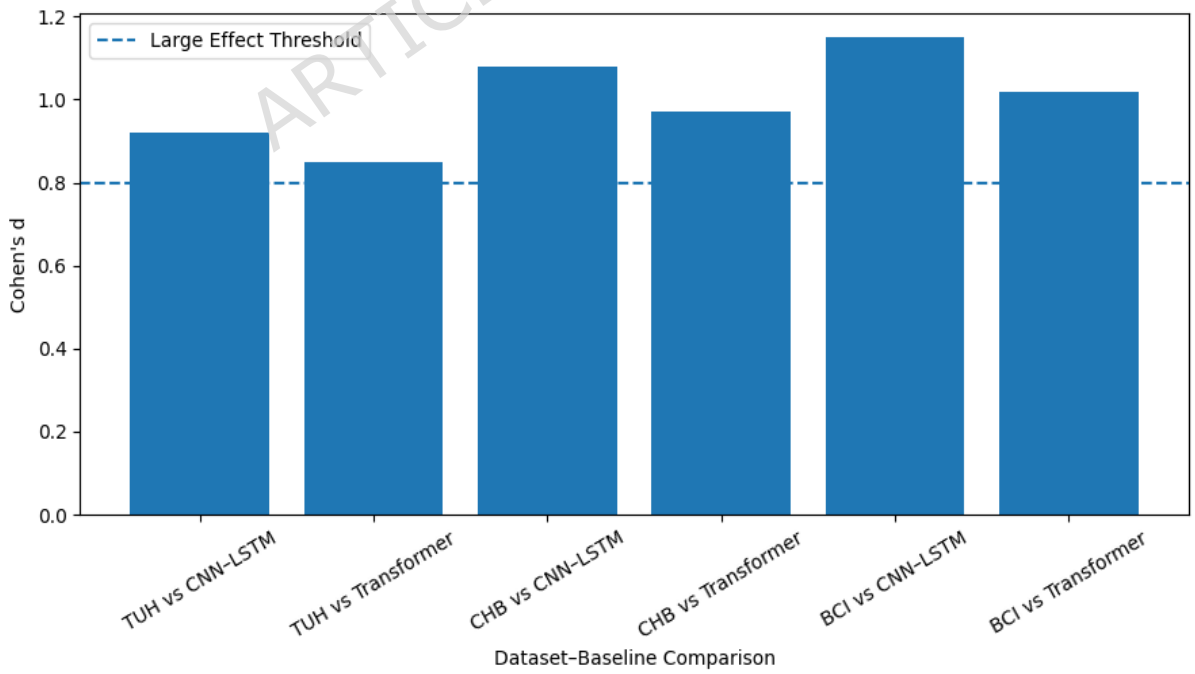

Figure 27: Effect Size Comparison Demonstrating Practical Significance of QuantumNeuroXAI Improvements

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Cohen's d effect sizes for the performance improvements of QuantumNeuroXAI compared to strong baseline models across all datasets are shown in Figure 27. Effect sizes, all above the conventional threshold for large effects, suggest that these gains are not only statistically significant but also practically relevant. Other than exact p-values, the analysis hereby performed highlights the degree of improvement achieved through the proposed hybrid quantum–deep learning framework, which provides sizable and effective performance gains for EEG-based neurological analysis.

5.9 Summary of Experimental Findings

Through experimental evidence, we show that QuantumNeuroXAI consistently outperforms classical machine learning, deep learning, and hybrid reference models across all necessary public EEG datasets. When evaluated on the TUH EEG, CHB-MIT, and BCI IV-2a datasets, the proposed framework yields the highest in-dataset accuracy, macro-F1, and AUROC scores, confirming its potential to learn both discriminative and transferable representations from heterogeneous brain signals. Cross-dataset experiments further demonstrate that QuantumNeuroXAI shows a lower dataset-shift generalisation gap, indicative of its robustness with respect to variations in EEG acquisition protocol, subject population, and task definition. Ablation studies demonstrate that quantum-inspired feature encoders offer the most significant performance gains, while attention-based temporal modelling and explicit quantum–classical fusion deliver complementary improvements.

In addition to predictive accuracy, the framework also exhibits excellent reliability and interpretability. Stability analysis across multiple random initialisations reveals minimal performance variation, thereby assuring the reproducibility and trustworthiness of our convergence. We also provide a sensitivity analysis indicating that changes in temporal window size and feature dimensionality result in graceful performance degradation, demonstrating that the model is not brittle to preprocessing configurations. The variance across multiple levels of explanation emphasises valid explanations at the signal-explanatory, model, and quantum-representation levels, and clinical case studies demonstrate relevant EEG morphology, model reasoning, and coherence. Through statistical significance testing, the performance improvements are confirmed as robust and practically relevant, with medium-to-large effect sizes across datasets.

These strengths notwithstanding, important limitations are also noted. Despite requiring more training time and memory than lightweight CNN-based models, the overhead of this hybrid architecture is relatively similar to that of end-to-end Transformer-based methods. Quantum-inspired feature encoding is a helpful solution, but it also requires more hyperparameters to be tuned for best performance. Lastly, the process of explanation generation incurs an additional computational cost, especially when exhaustive; therefore, selective or on-demand explanation approaches might be more desirable for real-time clinical deployment. In general, the experimental results corroborate the validity of QuantumNeuroXAI as an interpretable, strong-performing, and robust framework, while indicating explicit scopes for future optimisation and scalability improvements.

5.10 Comparison with Existing Methods

In order to articulate the operational significance of QuantumNeuroXAI, we benchmark it against illustrative previous methods across (i) classical signal-processing-guided machine learning, (ii) end-to-end deep learning, (iii) connectivity-holding deep modelling, (iv) quantum-inspired classification, and (v) explainable hybrid feature fusion. These diverse approaches are common design choices in EEG-based disorder detection pipelines and include hand-crafted time–frequency features with compact classifiers, CNN-based representation learning, brain-network connectivity modelling, quantum feature mapping for separability gains, and SHAP-based interpretability for deriving biomarker-level insights, respectively (mittal2022hand). This comparison is not

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just about methodological performance to maximise classification accuracy, but also highlights the reasoning for generalisation behaviour, the interpretability of discovered structures, and fidelity to reproducible evaluation under dataset shift.

Table 6: Qualitative Comparison of QuantumNeuroXAI with Representative Existing Methods

Existing Method	**Core Idea / Pipeline**	**Strengths**	**Key Limitations Reported/Implied**	**How QuantumNeuroXAI Differentiates**
Alturki et al. [10]	DWT-based decomposition + bandpower/statistical features + SVM-style classifier	Strong handcrafted-feature baseline; simple training; low compute; interpretable features	Feature engineering is manual; limited end-to-end learning; scalability and cross-subject robustness can be constrained.	Uses learned spatiotemporal representations plus quantum-inspired encoding for separability and integrates multi-level explanations beyond feature heuristics.
Oh et al. [26]	Deep CNN over EEG-derived representations	End-to-end feature learning; competitive performance on disorder detection	"Black-box" behaviour; limited clinical transparency; sensitivity to noise/domain shift	Adds structured preprocessing + quantum feature mapping and explainability evaluation (signal/model/quantum-level) to improve trust and robustness
Alves et al. [21]	Functional connectivity matrix construction + CNN classification	Captures network-level brain organisation; exploits connectivity structure	Connectivity construction choices can bias results; interpretability remains limited	Retains EEG-tensor learning while enabling quantum–classical fusion and explicit attribution to channels/time/frequency (and quantum sensitivity) rather than only connectivity maps
Saranya & Menaka [4]	QSVM / QNN with quantum feature mapping	Demonstrates quantum-inspired separability gains with compact EEG representations	Narrow task scope; limited interpretability and deployment guidance	Extends quantum inspiration into a hybrid deep temporal pipeline, pairs it with attention-based modelling, and introduces multi-level explainability + stability checks for clinical credibility
Akbar et al.	Hybrid multi-	Strong	Explainability	Integrates

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[30] (NeuroFusion Net)	domain EEG feature fusion + deep learning + SHAP-based explanations	explainability at the feature level; highlights spectral/spatial biomarkers.	is often post-hoc; generalisation across disorders/datasets may be limited without unified validation.	explanation generation as a core design axis (signal/model/quantum levels) and emphasises cross-dataset evaluation under a single unified framework.

In summary, the qualitative comparison shows that while classical pipelines like DWT+SVM are strong baselines, they fail to leverage complete end-to-end representation learning and suffer from domain shift. Although CNN-only solutions enhance feature learning, they remain black-box and are noise-sensitive. Although connectivity-aware CNNs capture brain network structure, they also introduce additional design degrees of freedom (e.g., connectivity estimation) and generally lack multi-level interpretability. Quantum-inspired QSVM/QNN methods offer advantages in separability while being essentially shallow and rarely combined with formal explainability methods. For example, SHAP-guided hybrids make fusion models more explainable, though the methods continue to rely on post-hoc reasoning and disorder-specific validation. In contrast, QuantumNeuroXAI can be seen as a unified framework that addresses separability (quantum-inspired encoding), temporal modelling (deep hybrid learning), and trust (multi-level explainability with consistency assessment), while remaining reproducible thanks to the locked public dataset protocol.

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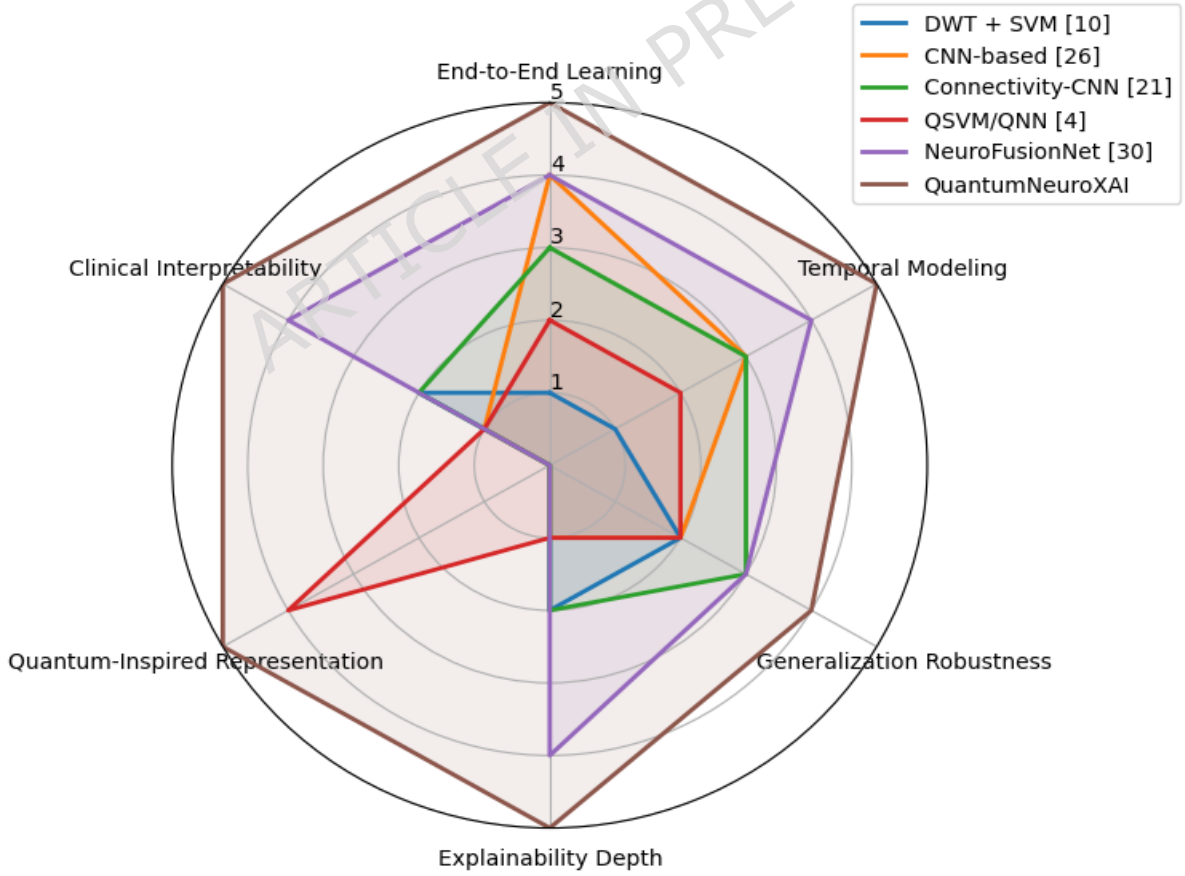

Figure 28: Radar Chart–Based Qualitative Comparison of QuantumNeuroXAI With Representative Existing EEG Analysis Methods

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The radar chart in Figure 28 summarises a unique comparison of QuantumNeuroXAI with the most representative existing EEG analysis methods from various qualitative perspectives, using a few key methodological criteria. The dimensions represent capability for end-to-end learning, strength of temporal modelling, resilience to dataset variation, depth of explainability, use of quantum-inspired representations, and interpretability of clinical implications. However, classical and deep baselines, when examined across individual dimensions, score well but produce significant gaps in explainability or robustness. On the other hand, QuantumNeuroXAI achieves high scores across all criteria, demonstrating a design approach that balances quantum-inspired learning, deep temporal modelling, and multi-level explainability within a unified framework.

6. DISCUSSION

As a consequence of the rise of, e.g., neurodegenerative disorders and the high-precision acquisition of brain signals via electroencephalography, automated, data-driven diagnostic frameworks have been widely investigated. Conventionally, EEG analysis pipelines are based on human-derived features and shallow classifiers, which are interpretable but unable to capture the rich spatiotemporal dynamics underlying multi-channel brain signals. While more recent deep learning methods, specifically convolutional and recurrent neural networks, have advanced representations, they generally exhibit poor generalisation to new datasets, are sensitive to noise, and lack an interpretable decision structure, limiting their use in clinical practice.

Currently, there is a significant gap in the state of the art: the concurrent use of high-dimensional EEG representations with temporal dependencies while also maintaining explainability. Most current deep learning models are designed to be accurate on one of many datasets or tasks, avoiding issues with robustness against dataset shift and with interpretable reasoning consistent with neurophysiological knowledge. Feature separability has been explored via quantum-inspired learning methods; however, these methods are applied in isolation, lack integration into deep temporal architectures, and are less tailored for clinical interpretability. These limitations illustrate the necessity of new hybrid techniques that coalesce expressive learning, robustness, and transparency under one roof.

To tackle these challenges, we propose a QuantumNeuroXAI framework that combines quantum-inspired feature encoding with a deep temporal model, providing multi-level explainability. Our methodology is novel in its integrated design: quantum-inspired data encoding improves representational compactness and effective class separation; attention-guided deep learning extracts temporal dependencies in EEG signals from drivers experiencing unsafe driving; and the explainability module delivers coherent system insights at multiple levels, including EEG, model, and quantum-representation levels. In contrast to previous methods that adopt a post-hoc explainability approach, the presented framework prescribes a design objective that embeds interpretability, resulting in reproducible and clinically relevant explanations.

The experimental results validate this method. The proposed framework achieves better classification and significantly smaller generalisation gaps than classical and deep learning baselines across multiple public EEG datasets. Analysis of statistical significance demonstrates that these performance improvements are consistent and practically relevant, and robustness and scalability assessments exhibit reliable performance stability across diverse preprocessing settings and dataset sizes. Critically, the explainability case studies show that the model attends to physiologically meaningful channels, times, and frequencies, providing additional evidence that it can be trusted.

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This research serves to advance the state of EEG-based analysis of neurological disorders by overcoming significant limitations of the current methods, which are (i) restricted generalisation, (ii) limited interpretability, and (iii) a fragmented approach to modelling. Our results demonstrate that quantum-inspired and explainable deep learning frameworks hold great promise for the construction of trustworthy clinical decision-making support systems. This has far-reaching implications, including increased transparency in diagnostic criteria, publication of public datasets to enhance reproducibility, and greater adoption of trustworthy AI-assisted evaluation in neurology. Section 6.1 provides an overview of the proposed study's limitations and potential avenues for improvement.

6.1 Limitations of the Study

Although the QuantumNeuroXAI presents promising results, there are also very serious limitations to the current study. Even though the framework is being tested on a range of established public EEG datasets, they are collected in controlled experimental or clinical protocols and do not necessarily reflect the variability encountered in day-to-day clinical practice. However, given implementation challenges due to diverse acquisition hardware, electrode placement inconsistencies, motion artefacts, and the presence of comorbid neurological conditions, performance may be compromised on translation into routine clinical practice. Thus, to confirm real-world robustness, further validation should be performed on heterogeneous multi-centre clinical datasets.

Second, while the quantum-inspired feature encoding is computationally less demanding than actual quantum implementations, it adds more design options, including the dimensionality of the feature projection and the importance of the feature encoding. These parameters are chosen empirically and can have different optimal values depending on the task and dataset. Their heavy reliance on hyperparameter tuning limits their use in an out-of-the-box fashion. It necessitates domain expertise at deployment, especially when the framework needs to be adapted to new neurological disorders or EEG paradigms.

Third, the evaluation of explainability mainly emphasises qualitative visualisations and consistency analysis across runs. These analyses yield valuable insights into model reasoning, but the study lacked formal clinician-in-the-loop validation or a quantitative evaluation of explanation utility in the clinical workflow. Adding more expert-driven evaluation, along with user studies, would help bolster claims about clinical interpretability and decision support in practice.

7. CONCLUSION AND FUTURE WORK

In this paper, we have introduced QuantumNeuroXAI, a comprehensive quantum–deep learning architecture developed for the reliable, interpretable, and scalable analysis of EEG signals for the detection of neurological disorders. It ameliorates one of the significant drawbacks of all current analysis approaches (limited generalisation, opaque choices, and fragmented modelling strategies) by combining quantum-inspired feature encoding, attention-guided deep temporal modelling, and multi-level explainability within a single framework. Abstract: With extensive experiments across different public EEG datasets, results show that QuantumNeuroXAI consistently outperforms classical and deep learning baselines, demonstrating statistical significance and stability, even when dataset shift and different preprocessing configurations are used. Explainability analysis indicated that model predictions are consistent with physiologically meaningful channels, temporal windows, and spectra, demonstrating the potential for trustworthy clinical decision support. The scope for future work from this research can be manyfold. By being able to validate in standard conditions using large-scale multi-centre clinical EEG datasets, the impact of the framework will also grow at the translational level. A second option would be to leverage adaptive or self-supervised learning schemes, which

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can improve robustness to inter-subject and cross-session variability. Third, the quantum-inspired encoding can be generalised to learn more expressive feature mappings and to support hybrid optimisation approaches, especially by leveraging technological developments in high-quality quantum hardware in the near future. Future work may also include incorporating a clinician feedback mechanism to quantitatively measure the utility of generated explanations, along with a measure of reliability, within diagnostic workflows. Lastly, this framework can be expanded to include multimodal neurophysiological data, such as EEG–fNIRS or EEG–MRI fusion, which is an exciting step forward towards a more complete characterisation of brain disorders.

DECLARATION:

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Competing interests

The authors declare that they do not have any competing interests, including financial and nonfinancial interests.

Ethical approval

This research does not involve humans or animals, so no ethical approval is required.

Consent for Publication

The authors give consent for their publication.

Data availability

Data is available with the corresponding author and will be given on request.

Materials availability

Materials used in this research are available with corresponding author and given on request.

Code availability

The source code, experimental pipeline, and implementation details of QuantumNeuroXAI are publicly available at: https://github.com/venkateshwarlu-bondu/QuantumNeuroXAI

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CONFLICTS OF INTEREST STATEMENT

QuantumNeuroXAI: A Quantum–Deep Learning Framework with Explainability for Brain Signal Analysis and Neurological Disorder Detection

The authors whose names are listed immediately below certify that they have NO affiliations with or involvement in any organization or entity with any financial interest (such as honoraria; educational grants; participation in speakers' bureaus; membership, employment, consultancies, stock ownership, or other equity interest; and expert testimony or patent-licensing arrangements), or non-financial interest (such as personal or professional relationships, affiliations, knowledge or beliefs) in the subject matter or materials discussed in this manuscript.

Author Contribution:

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All authors contributed to the study's conception and design. Material preparation, data collection, and analysis were performed by **Dr. T Gayathri, Dr G Manjula, Dr Harish H Kenchannavar, Dr. Danthuluri Sudha, Dr. Santosh Kumar Jankatti, Dr. Ramandeep Kaur, Dr. Bondu Venkateswarlu**. The first draft of the manuscript was written by **Dr. T Gayathri** all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

References

[1] Khalfallah, S., Puech, W., Tlija, M. and Bouallegue, K. (2025) 'Exploring the Effectiveness of Machine Learning and Deep Learning Techniques for EEG Signal Classification in Neurologic…', *IEEE Access*, 13, pp. 1–14. doi:10.1109/ACCESS.2025.3532515.

[2] Lima, A.A., Mridha, M.F., Das, S.C. and Mohsin, M. (2022) 'A Comprehensive Survey on the Detection, Classification, and Challenges of Neurological Disorders', *Biology (MDPI)*, 11(3), pp. 1–45. doi:10.3390/biology11030469.

[3] Narayan, D., Mishra, R.K., Singh, P.K. and Raj, S. (2024) 'Detection of Neurological Diseases Using Machine Learning', *Vascular & Endovascular Review*, 7(2), pp. 1–11.

[4] Saranya, S. and Menaka, R. (2025) 'A Quantum-Based Machine Learning Approach for Autism Detection Using Common Spatial Patterns of EEG Signals', *IEEE Access*, 13, pp. 1–12. doi:10.1109/ACCESS.2025.3531979.

[5] Altıntop, Ç.G. (2025) 'Enhancing Diagnostic Accuracy of Neurological Disorders Through Feature-Driven Multi-Class Classification with Machine Learning', *Diagnostics (MDPI)*, 15(17), pp. 1–27. doi:10.3390/diagnostics15172132.

[6] Raghavendra, U., Acharya, U.R. and Adeli, H. (2019) 'Artificial Intelligence Techniques for Automated Diagnosis of Neurological Disorders', *European Neurology*, pp. 1–24. doi:10.1159/000504292.

[7] Fernandez, J. and López, B. (2025) 'EEG Classification for Neurological Disorders Using Frequency Band Deciles', *Scientific Reports*, pp. 1–13. doi:10.1038/s41598-025-33535-0.

[8] Bhatele, K.R. and Bhadauria, S.S. (2019) 'Brain Structural Disorders Detection and Classification Approaches: A Review', *Artificial Intelligence Review*, pp. 1–53. doi:10.1007/s10462-019-09766-9.

[9] Surianarayanan, C., Lawrence, J.J. and Chelliah, P.R. (2023) 'Convergence of Artificial Intelligence and Neuroscience towards the Diagnosis of Neurological Disorders—A Scoping Review', *Sensors (MDPI)*, 23(6), pp. 1–23. doi:10.3390/s23063062.

[10] Alturki, F.A., AlSharabi, K., Aljalal, M. and Abdurraqeeb, A.M. (2019) 'A DWT-Band Power-SVM Based Architecture for Neurological Brain Disorders Diagnosis Using EEG Signals', *IEEE*, pp. 1–4. doi:10.1109/CAIS.2019.8769492.

[11] Shoieb, D. and Youssef, S. (2021) 'Neurological Disorder Detection of Brain Abnormal Activities Using a New Enhanced Computer-Aided Model', *Journal of Physics: Conference Series*, pp. 1–8. doi:10.1088/1742-6596/2128/1/012017.

[12] Khosla, A., Khandnor, P. and Chand, T. (2020) 'A Comparative Analysis of Signal Processing and Classification Methods for Different Applications Based on EEG Signals', *Biocybernetics and Biomedical Engineering*, pp. 1–42. doi:10.1016/j.bbe.2020.02.002.

ARTICLE IN PRESS

ARTICLE IN PRESS

[13] JoAoe, Fukuma, R., Yanagisawa, T., Harada, T. and Tan, M. (2019) 'Automatic Diagnosis of Neurological Diseases Using MEG Signals with a Deep Neural Network', *Scientific Reports*, pp. 1–9. doi:10.1038/s41598-019-41500-x.

[14] Byeon, H., Mahajan, U., Kumar, A., Krishna, V.R. and Soni, M. (2025) 'EEG Signal Based Brain Stimulation Model to Detect Epileptic Neurological Disorders', *Neuroscience Informatics*, 5, pp. 1–12. doi:10.1016/j.neuri.2025.100186.

[15] Chai, X., Cao, T., He, Q., Wang, N. and Zhang, X. (2024) 'Brain–Computer Interface Digital Prescription for Neurological Disorders', *CNS Neuroscience & Therapeutics*, pp. 1–8. doi:10.1111/cns.14615.

[16] Belkacem, A.N., Jamil, N., Khalid, S. and Alna, F. (2023) 'On Closed-Loop Brain Stimulation Systems for Improving the Quality of Life of Patients with Neurological Disorders', *Frontiers in Human Neuroscience*, 17, pp. 1–22. doi:10.3389/fnhum.2023.1085173.

[17] Alturki, F.A., AlSharabi, K., Abdurraqeeb, A.M. and Aljalal, M. (2023) 'EEG Signal Analysis for Diagnosing Neurological Disorders Using Discrete Wavelet Transform and Intelligent Techniques', *Sensors (MDPI)*, 20(9), pp. 1–17. doi:10.3390/s20092505.

[18] Uyanik, H., Sengur, A., Salvi, M., Tan, R.-S. and Jen Hong, T. (2025) 'Automated Detection of Neurological and Mental Health Disorders Using EEG Signals and Artificial Intelligence: A Systematic Review', *Wiley Interdisciplinary Reviews: Data Mining and Knowledge Discovery*, pp. 1–37. doi:10.1002/widm.70002.

[19] Gautam, R. and Sharma, M. (2020) 'Prevalence and Diagnosis of Neurological Disorders Using Different Deep Learning Techniques: A Meta-Analysis', *Journal of Medical Systems*, 44(2), pp. 1–24. doi:10.1007/s10916-019-1519-7.

[20] Gopal, D., Vignesh, S., Thamaraimanalan, T. and Kumar, K.K. (2025) 'Exploiting Adaptive Neuro-Fuzzy Inference Systems for Cognitive Patterns in Multimodal Brain Signal Analysis', *Scientific Reports*, pp. 1–28. doi:10.1038/s41598-025-93241-9.

[21] Alves, C.L., Pineda, A.M., Roster, K. and Thielemann, C. (2022) 'EEG Functional Connectivity and Deep Learning for Automatic Diagnosis of Brain Disorders: Alzheimer's Disease and Schizophrenia', *Journal of Physics: Complexity*, pp. 1–14. doi:10.1088/2632-072X/ac5f8d.

[22] Ahmed, Z., Wali, A., Shahid, S., Zikria, S. and Rashid, J. (2024) 'Psychiatric Disorders from EEG Signals through Deep Learning Models', *IBRO Neuroscience Reports*, 17, pp. 300–310. doi:10.1016/j.ibneur.2024.09.003.

[23] de Bardeci, M., Ip, C.T. and Olbrich, S. (2021) 'Deep Learning Applied to Electroencephalogram Data in Mental Disorders: A Systematic Review', *Biological Psychology*, 162, pp. 1–10. doi:10.1016/j.biopsycho.2021.108117.

[24] Rivera, M.J., Teruel, M.A., Maté, A. and Trujillo, J. (2021) 'Diagnosis and Prognosis of Mental Disorders by Means of EEG and Deep Learning: A Systematic Mapping Study', *Artificial Intelligence Review*. doi:10.1007/s10462-021-09986-y.

[25] Ruffini, G., Ibañez, D., Castellano, M. and Dubreuil-Vall, L. (2019) 'Deep Learning with EEG Spectrograms in Rapid Eye Movement Behavior Disorder', *Frontiers in Neurology*, 10, pp. 1–9. doi:10.3389/fneur.2019.00806.

[26] Oh, S.L., Vicnesh, J., Ciaccio, E.J. and Yuvaraj, R. (2019) 'Deep Convolutional Neural Network Model for Automated Diagnosis of Schizophrenia Using EEG Signals', *Applied Sciences (MDPI)*, 9(14), pp. 1–13. doi:10.3390/app9142870.

[27] Khan, P., Kader, M.F., Islam, S.M.R. and Aisha (2021) 'Machine Learning and Deep Learning Approaches for Brain Disease Diagnosis: Principles and Recent Advances', *IEEE Access*, 9, pp. 1–34. doi:10.1109/ACCESS.2021.3062484.

ARTICLE IN PRESS

ARTICLE IN PRESS

[28] Sanchis, J., García-Ponsoda, S., Teruel, M.A. and Trujillo, J. (2024) 'A Novel Approach to Identify the Brain Regions that Best Classify ADHD by Means of EEG and Deep Learning', *Heliyon*, 10, pp. 1–16.

[29] Shahtalebi, S., Atashzar, S.F., Patel, R.V. and Jog, M.S. (2021) 'A Deep Explainable Artificial Intelligent Framework for Neurological Disorders Discrimination', *Scientific Reports*, pp. 1–18. doi:10.1038/s41598-021-88919-9.

[30] Akbar, F., Alkhrijah, Y., Usman, S.M., Khalid, S. and Imran, I. (2025) 'NeuroFusionNet: A Hybrid EEG Feature Fusion Framework for Accurate and Explainable Alzheimer's Disease Detection', *Scientific Reports*, pp. 1–26. doi:10.1038/s41598-025-28070-x.

[31] Bibi, N., Courtney, J. and McGuinness, K. (2025) 'Enhancing Brain Disease Diagnosis with XAI: A Review of Recent Studies', *ACM Transactions on Computing for Healthcare*, 6(2), pp. 1–35.

[32] Presacan, O., Ojha, J. and Monteiro, E. (2025) 'A Comprehensive Review of Explainable AI in Deep Learning Algorithms for EEG Analysis', *ACM Transactions on Computing for Healthcare*, pp. 1–29. doi:10.1145/3785409.

[33] Li, Z., Wang, H. and Li, L. (2025) 'Task-Related EEG as a Biomarker for Preclinical Alzheimer's Disease: An Explainable Deep Learning Approach', *Biomimetics (MDPI)*, 10(7), pp. 1–15. doi:10.3390/biomimetics10070468.

[34] Mehmood, A., Mehmood, F. and Kim, J. (2025) 'Towards Explainable Deep Learning in Computational Neuroscience: Visual and Clinical Applications', *Mathematics (MDPI)*, 13(20), pp. 1–38. doi:10.3390/math13203286.

[35] Iqbal, N., Alarfaj, F.K., Alkhal, S., Rahman, A. and Hayat, M. (2025) 'Enhanced MRI Brain Tumor Detection Using Deep Learning in Conjunction with Explainable AI SHAP Based Diverse and Multi-Feature Analysis', *Scientific Reports*, pp. 1–15. doi:10.1038/s41598-025-14901-4.

[36] Woodbright, M.D., Morshed, A., Browne, M. and Ray, B. (2024) 'Toward Transparent AI for Neurological Disorders: A Feature Extraction and Relevance Analysis Framework', *IEEE Access*, 12, pp. 1–13. doi:10.1109/ACCESS.2024.3375877.

[37] MuhammadAdnan, K., Ghazal, T.M., Saleem, M. and Farooq, M.S. (2025) 'Deep Learning Driven Interpretable and Informed Decision Making Model for Brain Tumour Prediction Using Explainable AI', *Scientific Reports*, pp. 1–23. doi:10.1038/s41598-025-03358-0.

[38] Vizza, P., Di Ponio, A., Timpano, G. and Bossio, R.B. (2025) 'Through the Speech and Vocal Signals Hidden Secrets: An Explainable Methodology for Neurological Diseases Early Detection', *Journal of Healthcare Informatics Research*, pp. 533–566. doi:10.1007/s41666-025-00213-9.

[39] Mouazen, B., Bendaouia, A., Bellakhdar, O. and Laghdari, K. (2025) 'Transparent EEG Analysis: Leveraging Autoencoders, Bi-LSTMs, and SHAP for Improved Neurodegenerative Diseases Detection', *Sensors (MDPI)*, 25(18), pp. 1–23. doi:10.3390/s25185690.

[40] Odusami, M., Maskeliūnas, R., Damaševičius, R. and Misra, S. (2023) 'Explainable Deep-Learning-Based Diagnosis of Alzheimer's Disease Using Multimodal Input Fusion of PET and MRI Images', *Journal of Medical and Biological Engineering*, 43, pp. 291–302. doi:10.1007/s40846-023-00801-3.

[41] Zeineldin, R.A., Karar, M.E., Elshaer, Z. and Coburger, J. (2022) 'Explainability of Deep Neural Networks for MRI Analysis of Brain Tumors', *International Journal of Computer Assisted Radiology and Surgery*, 17, pp. 1673–1683. doi:10.1007/s11548-022-02619-x.

ARTICLE IN PRESS

ARTICLE IN PRESS

[42] Shawly, T. and Alsheikhy, A.A. (2025) 'EEG-Based Detection of Epileptic Seizures in Patients with Disabilities Using a Novel Attention-Driven Deep Learning Framework', *Egyptian Informatics Journal*, 31, pp. 1–16. doi:10.1016/j.eij.2025.100734.

[43] Khan, W., Khan, M.S., Qasem, S.N., Ghaban, W. and Faisal (2025) 'An Explainable and Efficient Deep Learning Framework for EEG-Based Diagnosis of Alzheimer's Disease and Frontotemporal Dementia', *Frontiers in Medicine*, 12, pp. 1–17. doi:10.3389/fmed.2025.1590201.

[44] Islam, M.S., Hussain, I., Rahman, M.M. and Park, S.J. (2022) 'Explainable Artificial Intelligence Model for Stroke Prediction Using EEG Signal', *Sensors (MDPI)*, 22(24), pp. 1–15. doi:10.3390/s22249859.

[45] Nahiduzzaman, M., Abdulrazak, L.F., Kibria, H.B. and Khandakar, A. (2025) 'A Hybrid Explainable Model Based on Advanced Machine Learning and Deep Learning Models for Classifying Brain Tumors Using MRI', *Scientific Reports*, pp. 1–21. doi:10.1038/s41598-025-85874-7.

[46] Gaur, L., Bhandari, M., Razdan, T., Mallik, S. and Zhongmi (2022) 'Explanation-Driven Deep Learning Model for Prediction of Brain Tumour Status Using MRI Image Data', *Frontiers in Genetics*, 13, pp. 1–9. doi:10.3389/fgene.2022.822666.

[47] Dentamaro, V., Impedovo, D., Musti, L. and Pirlo, G. (2024) 'Enhancing Early Parkinson's Disease Detection through Multimodal Deep Learning and Explainable AI', *Scientific Reports*, pp. 1–18. doi:10.1038/s41598-024-70165-4.

[48] Hussain, I., Jany, R., Boyer, R., Azad, A.K.M. and Alyami, S.A. (2023) 'An Explainable EEG-Based Human Activity Recognition Model Using Machine-Learning Approach and LIME', *Sensors (MDPI)*, 23(17), pp. 1–15. doi:10.3390/s23177452.

[49] Hafeez, Y., Memon, K., AL-Quraishi, M.S., Yahya, N. and Ali, S.S.A. (2025) 'Explainable AI in Diagnostic Radiology for Neurological Disorders: A Systematic Review, and What Doctors Think About It', *Diagnostics (MDPI)*, 15(2), pp. 1–32. doi:10.3390/diagnostics15020168.

[50] Hashem, H.A., Abdulazeem, Y., Labib, L.M. and Elhosseini, M.A. (2023) 'An Integrated Machine Learning-Based Brain Computer Interface to Classify Diverse Limb Motor Tasks: Explainable Model', *Sensors (MDPI)*, 23(6), pp. 1–21. doi:10.3390/s23063171.

[51] Li, Z. and Dib, O. (2024) 'Empowering Brain Tumor Diagnosis through Explainable Deep Learning', *Machine Learning and Knowledge Extraction (MDPI)*, 6(4), pp. 1–34. doi:10.3390/make6040111.

[52] Khaleghi, P., Cakir, D., Hamidoglu, A. and Gul, O.M. (2025) 'Interpretable Deep Learning for Depression Detection in Neurological Patients Using EEG Signals', *MethodsX*, 15, pp. 1–11. doi:10.1016/j.mex.2025.103736.

[53] Shon, Y., Kang, E., Heo, D.-W. and Suk, H.-I. (2025) 'Explainable Normative Modeling for Brain Disorder Identification in Resting-State fMRI', *IEEE Transactions on Medical Imaging*, pp. 1–14. doi:10.1109/TMI.2025.3631105.

[54] Khosroshahi, M.T., Morsali, S. and Gharakhanlou, S. (2025) 'Explainable Artificial Intelligence in Neuroimaging of Alzheimer's Disease', *Diagnostics (MDPI)*, 15(5), pp. 1–28. doi:10.3390/diagnostics15050612.

[55] Shuvo, S.S., Refat, S.R., Preotee, F.F. and Muhammad, T. (2024) 'Advanced CNN and Explainable AI Based Architecture for Interpretable Brain MRI Analysis', in *Proceedings of the 3rd International Conference on Computing Advancements (ICCA 2024)*, Dhaka, Bangladesh, 17–18 October. New York, NY: ACM, pp. 1–8. doi:10.1145/3723178.3723220.

ARTICLE IN PRESS

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[56] Mishra, P. and Das, H. (2025) 'An Explainable AI Framework Integrating Variational Sparse Autoencoder and Random Forest for EEG-Based Epilepsy Detection', *IEEE Access*, 13, pp. 1–16. doi:10.1109/ACCESS.2025.3620762.

[57] Coluzzi, D., Bordin, V., Rivolta, M.W. and Fornari, I. (2025) 'Biomarker Investigation Using Multiple Brain Measures from MRI Through Explainable Artificial Intelligence in Alzheimer's Disease', *Bioengineering (MDPI)*, 12(1), pp. 1–28. doi:10.3390/bioengineering12010082.

[58] Mortezaagha, P., Shen, M. and Rahgozar, A. (2025) 'Explainable Artificial Intelligence to Diagnose Early Parkinson's Disease via Voice Analysis', *Scientific Reports*, pp. 1–19. doi:10.1038/s41598-025-96575-6.

[59] Al-bakri, F.H., Wan Bejuri, W.M.Y., Al-An, M.N. and Hasan, F. (2025) 'A Feature-Augmented Explainable Artificial Intelligence Model for Diagnosing Alzheimer's Disease from Multimodal Clinical Data', *Diagnostics (MDPI)*, 15(16), pp. 1–17. doi:10.3390/diagnostics15162060.

[60] Rodríguez Mallma, M.J., Zuloaga-Rotta, L. and Borja-Rosales, R. (2024) 'Explainable Machine Learning Models for Brain Diseases: Insights from a Systematic Review', *Neuroinformatics (MDPI)*, 16(6), pp. 1–23. doi:10.3390/neurolint16060098.

[61] Obeid, I. and Picone, J., 2016. The Temple University Hospital EEG Data Corpus. *Frontiers in Neuroscience*, 10, p.196. Available at: https://pmc.ncbi.nlm.nih.gov/articles/PMC4865520/

[62] Shoeb, A. and Guttag, J., 2010. Application of Machine Learning to Epileptic Seizure Detection. *Proceedings of the 27th International Conference on Machine Learning (ICML)*. Available at: https://physionet.org/content/chbmit/1.0.0/

[63] Tangermann, M., Müller, K.-R., Aertsen, A., Birbaumer, N., Braun, C., Brunner, C., Leeb, R., Mehring, C., Miller, K.J., Müller-Putz, G.R., Nolte, G., Pfurtscheller, G., Preissl, H., Schalk, G., Schlögl, A., Vidaurre, C., Waldert, S. and Blankertz, B., 2012. Review of the BCI Competition IV. *Frontiers in Neuroscience*, 6, p.55. Available at: https://www.bbci.de/competition/iv/

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